Project Report On

Ambarsthan Booking Management System

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Under the Guidance of
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In Partial fulfilment of
Bachelor of Technology
[B. Tech. Information Technology]
[2020-2021]
At



Department of Information Technology
Vishwakarma Institute of Information Technology, Pune 411048
Affiliated To



Savitribai Phule Pune University, Pune
Bansilal Ramnath Agarwal Charitable Trust's
AMBARSTHAN BOOKING MANAGEMENT SYSTEM

Vishwakarma Institute of Information Technology, Pune 411048



Certificate

This is to certify that the work entitled **"AMBARSTHAN BOOKING MANAGEMENT SYSTEM"** is a bonafide work carried out by Ms. Anisha Gandhi, Mr. Pratik Jagtap, Ms. Saloni Bonde, Mr. Vedant Bhatambrekar in partial fulfillment of the award of Bachelor of Technology in Information Technology, Savitribai Phule Pune University, Pune, during the year 2021. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the Bachelor of Technology Degree.

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Project Completion Certificate format by Industry

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To,
The Head-IT,
VIIT, Pune

We Tej Gyan Foundation, Pune, hereby certify that the following students of BTech-IT, VIIT have completed their final year BTech project titled: "**Proof of Concept (PoC)- Ambarsthan Booking management System**" at **Tej Gyan Foundation**, Pune (A Charitable Trust, registered under Bombay Public Trust Act, having its registered office at Vikrant Complex, Near Tapovan Mandir, Pimpri 411017) in the academic year 2020-2021.

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We thank them for the contribution and wish them good luck for the future.

For, Tej Gyan Foundation,

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Acknowledgement

We take this opportunity to thank Head of the Department Prof. P.Futane and our project guide Prof. S.G.Dedgaonkar for their valuable guidance and providing all the necessary facilities, which were indispensable in the completion of this project report. We are also thankful to all the staff members of the Department of Information Technology, VIIT, Pune for their valuable time, support, comments, suggestions and persuasion. We would also like to thank Mrs.Suruchi Dedgaonkar,Mrs.Jayashree Bagade,Mr.Rajendra Rode,TCS,TGF and the institute for providing the required infrastructure, internet and library facility.

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ABSTRACT

Tejgyan Foundation has its members across the globe. The Foundation conducts various Shivirs throughout the year. Recently, the foundation has built its Residential Complex “Ambarsthan”. As a further step, TGF wants an online accommodation management system for Ambarsthan. Project “Ambarsthan Management System” is initiated by TGF in collaboration with TCS and VIIT. The purpose of this project is to develop and implement an online accommodation management system for TGF, which will replace the manual method of booking for an accommodation. It focuses on minimizing human efforts and errors, thus providing efficiency in booking. The project follows Agile methodology. The tools and technologies used are: Spring Tool Suite for Spring Boot, MySQL Workbench for Database, Visual Studio Code for Angular, and Postman for API Servic

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Abreviations

- TGF : Tejgyan foundation
- ER : Entity Relationship
- ABMS - Ambarsthan Booking Management System
- BMS – Booking Management System

Chapter 1

Introduction

1.1 Motivation

Accommodation management software often becomes a great asset to the modern hospitality business. Still, lots of ready-made solutions offer complex interfaces, low functionality, and a bunch of unnecessary features. That's why more and more companies tend to develop software specifically for their business needs.
In our article, We are going to cover the main functions of the accommodation management system, explain what technologies are used for its development and explain why booking system that we are developing can run twice as effective with a top-rated system. Being a foundation , for which an accommodation system is built helps to reach out people based in several centres based in and out of India updating them with upcoming events, shivirs and booking accommodation for the same being located at their specific locations.

1.2 Need of Accommodation Management System

Without an online booking system, Khoji's will have to rely on phone calls and walk-ins only to make reservations. An online booking platform is a necessity. That's why we would like to point out the key benefits every good hotel booking system should contain, some benefits for using online booking management include :-

1. Work 24/7 - An online booking system works all the time. This gives freedom to potential Khoji's to book a room anytime they want. It also maximises sales/publicity because system dosen't tends to work in working hours. In fact, studies show that a 24/7 online reservation system greatly increases the number of hotel bookings.

2. Hassle-Free Management of Bookings - Hotel reservation systems will make TGF staff more efficient. They won't be tied to a phone waiting for guest calls. What's more, admin's will be able to take bookings around the clock, 7 days a week.

3. Get Smarter Insights - Usually, such systems come with various analytics tools. They can keep track of each reservation and all the related details. Thus, system will have all

the information to better understand your guests.

4. The Number of No-Shows Will Decrease - Prospective guests who reserve rooms online are much more likely to show up. And in the cases when someone doesn't show up, Ambarsthan online booking system will automatically free the reserved room, making it available for booking. Appointment reminders are also a great way to ensure that Khoji's will turn up.

5. Payments are Easier and Faster - Payments are also greatly eased with an online booking platform. System can require Khoji's to pay when they book and not worry about payments when they arrive. In addition, in case of a no-show, system can keep some of the money as compensation.

6. Cut TGF Workload - Online reservation systems reduce workloads for TGF's staff and optimize customer service. These platforms can make sure that bookings are synced and the availability is updated with each booking processing. A good hotel reservation system makes the work process carefree.

7.A New Booking Trend - The world is online now. Nowadays, customers use the Internet to discover everything they need to know. If you don't have an online presence, you are most likely missing out a great many opportunities. So, if you want to take advantage of the modern booking approach, it is a must to integrate your reservation platform with your product pages.

8.Meet the New Self-Service Portals - Online booking systems are designed to provide all the features of self-service portals. Potential Khoji's book services based on the open dates suggested by the system. Of course, TGF should have personalised customer care support to handle various queries. Still, with a well-maintained hotel reservation system, it will rarely need to deal with Khoji's directly.

1.3 Brief Introduction to Accommodation Management System

Online booking systems are designed to provide all the features of self-service portals. Potential guests book services based on the open dates suggested by the system. Of course, TGF should have personalised customer care support to handle various queries. Still, with a well-maintained hotel reservation system, TGF will rarely need to deal with customers directly. An online booking system developed to your needs will

simplify and improve your customer service, and will allow visualizing and assessing of the workload through optimizing processes and operations.An online booking system is a software solution that allows potential Khoji's to self-book and pay through Ambarsthan booking management website, and other channels, while giving you the best tools to run and scale your operation, all in one place. But if TGF'S Ambarsthan booking system has gotten far without an online booking system,it might think: does it even need one? What's with all the hype anyway? Ambarsthan is getting enough reservations already. Why bother taking on that extra expense? Booking systems can help you do way more than just passively accept bookings and payments online. Reservation technology (res-tech) has evolved to the point that it's become a central hub for managing every aspect of it's business, from marketing to distribution to operations. Now, TGF can track Ambarsthan's growth with real-time reporting. Avoid double bookings from online travel agents (OTAs) with channel management. Oversee all partnerships and commissions. Assign inventory and staff to individual activities. Get an idea of who's showing up and when. Break down your day-to-day priorities.

1.4 Reason behind Accommodation Management System

1. Far more secure than spreadsheets

2. Cuts TGF'S workload in half

3. Creates a better Khoji's experience

4. Helps stay organized

5. Prevents cancellations and no-shows

6. Smarter insights into TGF

1.5 Application

Booking Management System can also be applied to :

1. Bus reservation system

2.Tour Booking system

Chapter 2

Literature Survey

2.1 Literature Review

Table No 2.1: Comparison table of pros and cons of BMS

Booking Management System	Advantages	Disadvantages
1.	There is no limitation to booking hours of the system. Also, the system itself help's guide the facilities available.	It takes time to set-up.
2.	Business information is no longer confined to a desktop computer with a cloud based booking system.	You have to train your staff how to efficiently use the system.
3.	Manual data entry might be your one chance in the day to turn off your brain and listen to a podcast.	May cause huge traffic on site sometimes.
4.	Prevents cancellations and no-shows.	Should be organized enough : In case of same bookings.

2.2 Review of existing System

While brainstorming during system requirements and functionalities we came across various sites of booking management system .One of which was Airbnb ,which has efficient booking i.e when 2 users are booking at the same time ,the respective room is shown booked for 15 minutes even if the whole booking process till payment hasn't been done to avoid conflicts. Also, the disadvantage of the respective system is that they don't have update booking options which we have tried to implement in our system .Users of the system should be able to retrieve booking information of the respective shivir from the database. The types of rooms available, payment mode ,payment invoice ,booking

confirmation ,quick links, google map(route to the shivir) ,shivir timings, number of booked/available rooms, transportation facilities .The system will support two types of user privileges i.e Khoji(Shivir members) and employee(admin).Khoji's will have access to the Khoji functions and the employee's will have access to both the Khoji and booking management functions.

2.2.1 System Functions

Khoji Functions

- Book stay
- View Details
- Book transportation facilities
- Make payment
- Cancel Booking
- Update Booking
- Login / Register to the system
- Check bookings done
- Edit/Update/Delete guest information
- View his/her itinerary
- View his/her past booking details
- Location

User Functions

- Get all User information who have done the booking
- Get all types of rooms available
- View booking status of the respective user
- Calculate total sales
- Get timely updation/deletion updates of the respective booking
- View past events
- View past bookings

Admin Functions

- Add/Delete an event
- Add/Delete a booking
- Update charges

- Add new rooms
- Update dates of the events/bookings
- Number of servers/users logged in.

Chapter 3

Project Statement

3.1 Purpose behind the Project

Tejgyan Foundation has its members across the globe. The Foundation conducts various Shivirs throughout the year. Recently, the foundation has built its Residential Complex "Ambarsthan". As a further step, TGF wants an online accommodation management system for Ambarsthan. Project "Ambarsthan Management System" is initiated by TGF in collaboration with TCS and VIIT. The purpose of this project is to develop and implement an online accommodation management system for TGF, which will replace the manual method of booking for an accommodation. It focuses on minimizing human efforts and errors, thus providing efficiency in booking.

3.2 Decision of Scope

This project aims to develop a booking accommodation system which will include several functions like registering as a Khoji, registering as a guest, admin registeration for booking event and accommodation for self or on behalf of someone.This involves various purposes like gifted booking, special invitee and Vip boking.At the end of this project a interface is expected which will allow user to book accommodation and following event.

3.3 Methodology for solving this proposed theme

3.3.1 Proposed system Architecture

In our system, we are following this architecture. From TGF's global browser and mobile app the function is pinged into the system .The api's for the respective service is striked.The respective service stating the boking details or be it Khoji's details are therefore retrieved from the centralized database. Explained in the figure below

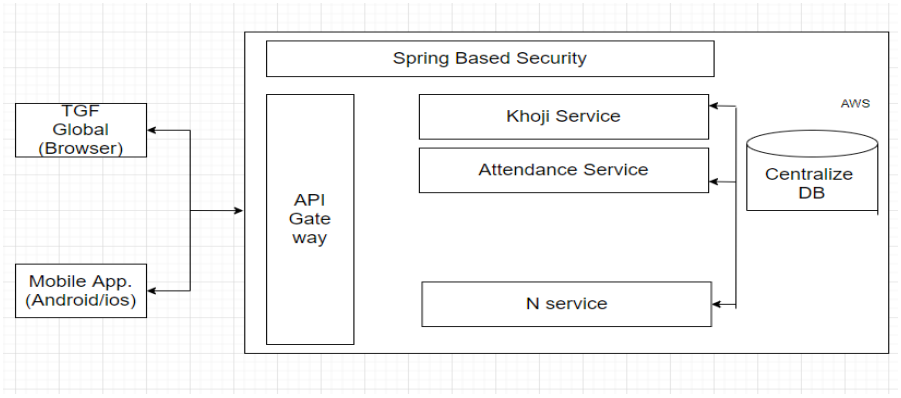


Fig 3.3.1: Proposed system Architecture

3.3.2 Working of the System

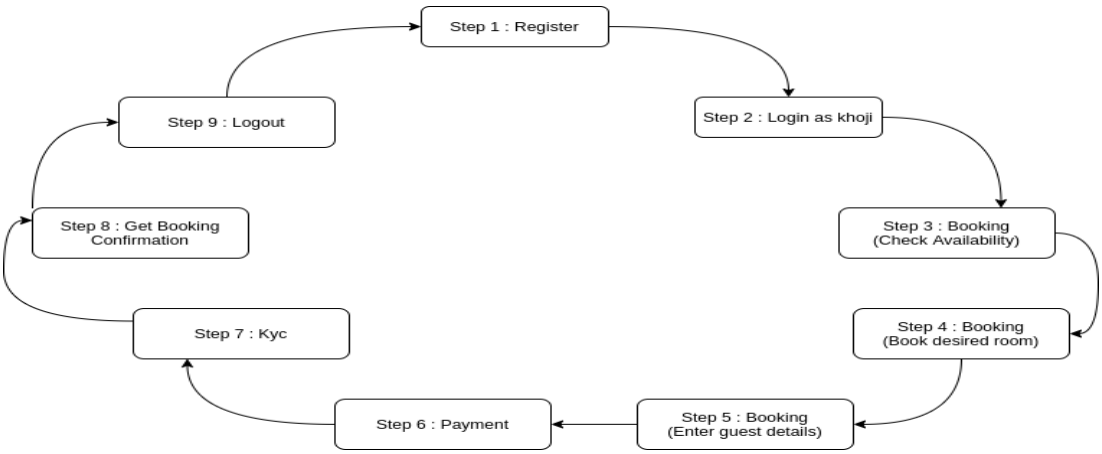


Fig 3.3.2: Process Flow

1. Register – User can get registered themselves as Khoji , guest Khoji ,temporary Khoji depending on their conditions

2. Login - Once registered , users have their own unique credentials to login to further access the facilities

3 .Booking - Based on user's check-in/check-out date through a specific login can further get access to the available accommodation

4. Booking (Accommodate) - Based on the availability in the previous step , user can take advantage of the same and book the desired type of accommodation based on availability

5 .Booking (Guest Details) - Enter user details for whom the accommodation has been booked

6. Payment - Do the online payment for the the type of accommodation and the number of the users for whom the bookings have been done

7. KYC - Do the required KYC

8. Get Booking Confirmation – Once completed with all the formalities, get your booking confirmation

9. Logout - If required, the user can log out of the system.

3.3.3 Flowchart of System

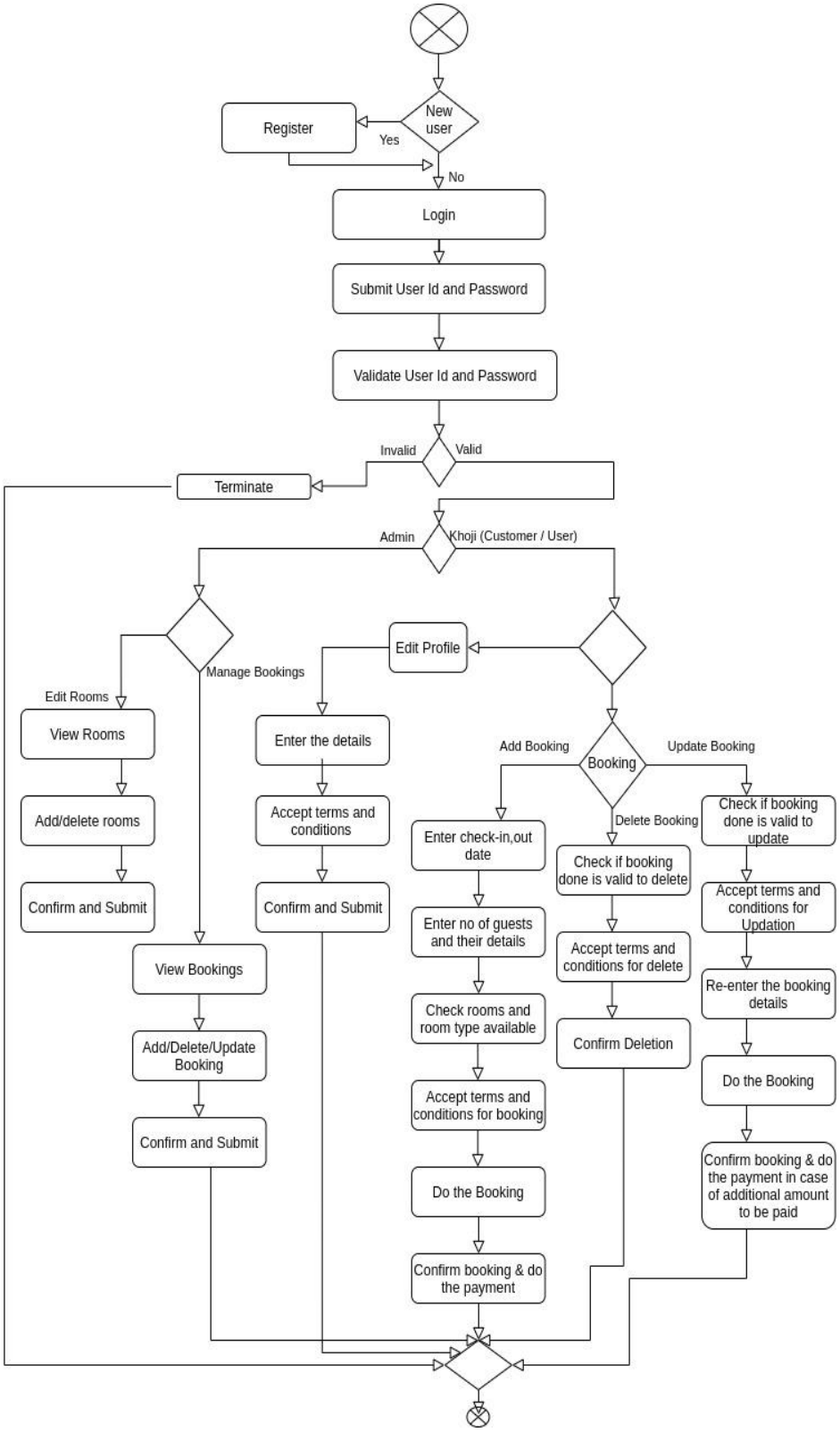


Fig 3.3.3 : Flowchart of system

Chapter 4

System requirement and specification

4.1 Software requirements specifications:

4.1.1 Introduction

Intended Audience and Reading Suggestions: Users of the system are staff , managers and members of TGF .The members are assumed to have basic knowledge of computer and internal browsing while the administrator should have more knowledge so that he/she can resolve small problems and perform information.

The intended audience includes all stakeholders as mentioned below

1. Developers
2. Users/Designers
3. Testers
4. Documentation Writers

4.1.2 User Classes and Characteristics

There are three users in the system:

User

- Go to the required site
- Register
- Login
- Book Accommodation
- Payment
- KYC

Admin

- Approve KYC
- Allot rooms
- Manage Gifted Booking
- Manage Special Invitee

- Manage bookings for VIP
- Edit Rooms
- Manage Bookings

System

- Perform the pre-processing
- Process unique ids for different users
- Save changes and update the database accordingly
- Don’t save a single booking for multiple users who functioned at the same time

4.1.3 Operating Environment

The system is a web based application that can work on any Operating System with a well equipped web browser. It just requires internet connection to book accommodation and avail all other facilities

4.1.4 External Interface Requirements

User Interfaces: User interface will be a web page which consists of a sign-in and sign-up option. Where users need to sign-up on the system before booking accommodation. Also there is a requirement field where users need to upload necessary documents and do online payment for the specific booking done by the user (if any).

Hardware Interfaces:

1. Client PC with Internet Connection.

Software Requirements :

Tools : Postman , framer , Workbench , Spring suite tool 4 , Microsoft Teams ,Jira, Visual Studio

4.1.5 Functional Requirement

- View rooms (Slide scroll)-When clicked on the specific room should show the details of the same and no other
- Google map - Accurate map from the users location to the shivir destination should be precise
- Login - New user id and password or incorrect when entered shouldn’t login into the system .Once logged in should open different UI depending on the user and admin.

- Register - All the requirements for login & password with the inculcated rules should be satisfied .If not ,the user entry for the same shouldn't be considered. New entry should be saved in the specific domain only i.e admin and user.
- Book rooms - Once clicked should open and book the selected room and for the payment the cost of the specific one should be displayed to pay and request for the same.

4.1.6 Other Non-functional Requirements

- Data Input Requirements:

1. Necessary documents to be uploaded for KYC should be scanned and be in specific size.

- Security Requirements:

1. System should not grant authentication to any unauthorised person. 2.
The system should not be vulnerable to the security attacks.
3. Information related to Admin password should be confidential.

- Software Quality Attributes:

The system ensures the following software quality attributes:

1. Reliability
2. Efficiency
3. Security
4. User-friendliness
5. Flexibility

4.1.7 Product Perspective

The product is supposed to be open source. It is a web based system implementing client-server model. The Ambarsthan booking management system provides a simple mechanism for users to book stays at the time of respective shivir and get views over feedbacks , payment options ,upcoming events and so.

- User account: The system allows the user to create their accounts in the system and provide features of updating and viewing profiles.
- Number of users being supported by the system: Though the number is precisely not mentioned but the system is able to support a large number of online users at a time.

- Search: search is simply a local search engine based on keywords.
- Booking Forum: Provides users with a platform to book their stay on the basis of reviews/feedback , types of rooms and cost.
- Ticketing system: Allows users to submit their issue to the admin in case his problems are not solved by FAQs and discussion forums.
- FAQs section: Frequently asked section contains answers to problems which are faced by the users.

4.1.8 Product Function

Some major product functionalities of the system are as follows:

- Booking Details : Types of rooms, no_of_rooms booked/available ,Check-in date, Check-out date ,number of guests, guest details ,location
- Khoji Description : name ,address ,mobile number, email
- Reservation Description : Khoji details , booking_Id, date of booking , date of travel

4.1.9 Modules – Developed for Business Logic

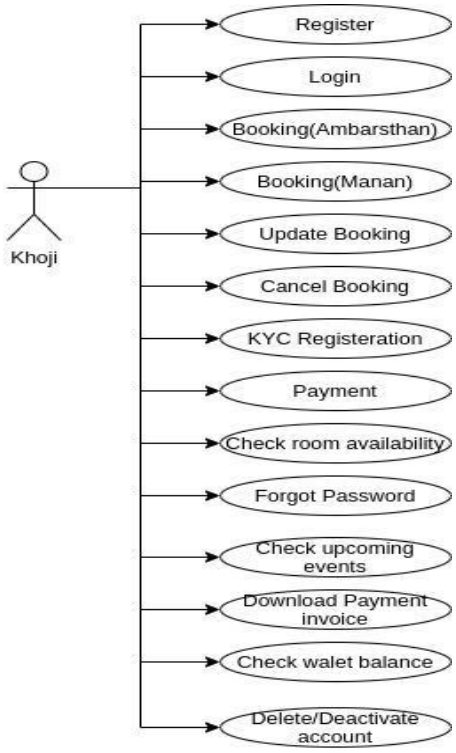
MODULES	PAGE
Guest Khoji Registration	Registration
Registration and Login	Registration & Login
Event Management	Home, Booking
Room availability Search	Booking
Accommodation Detail	Home
Gifted Booking	Booking
KYC	Booking
Room booking and payment	Booking
Google map + Location	All
Reminder	Home
Update Booking	Booking

MODULES	PAGE
Cancel Booking	Booking
FAQs	FAQ
Pickup Drop	Booking
Terms & Condition + Policies	Booking
Weather	Home
Ashram fees integration	Booking
Advertisement + Quotes	Home
Event Booking	Home
Wallet	Payment
Testimonial	Reviews
Customer history	History

Chapter 5

Project Analysis and Design

5.1 Use Case Diagram



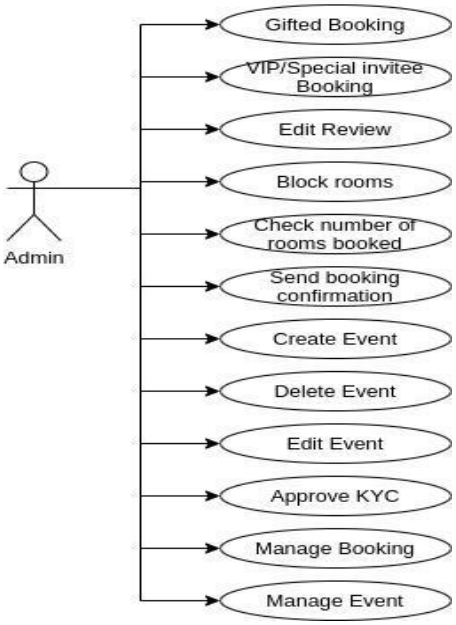
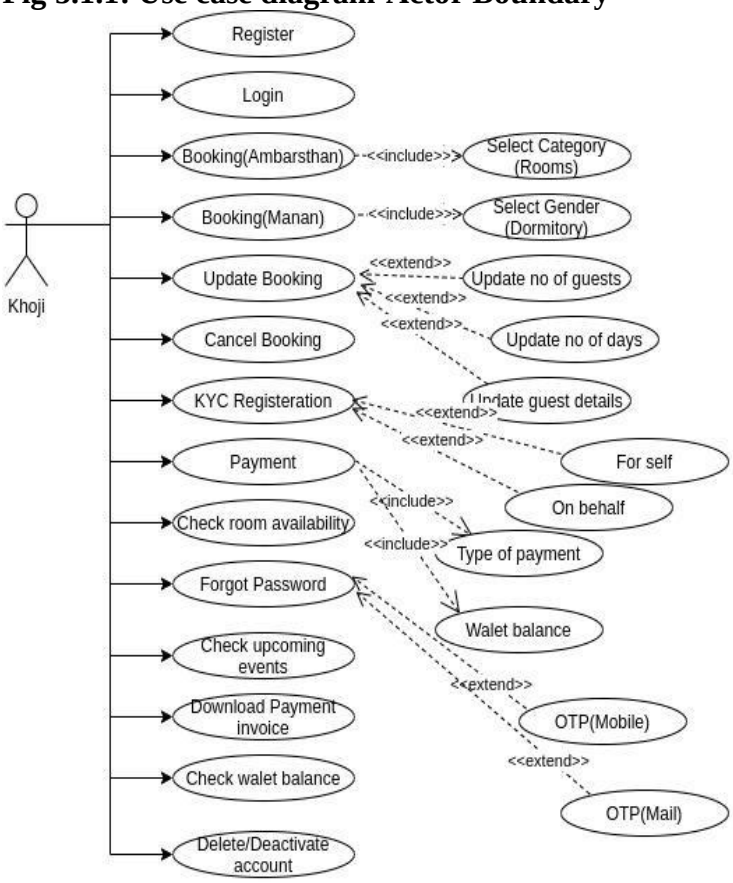


Fig 5.1.1: Use case diagram-Actor Boundary



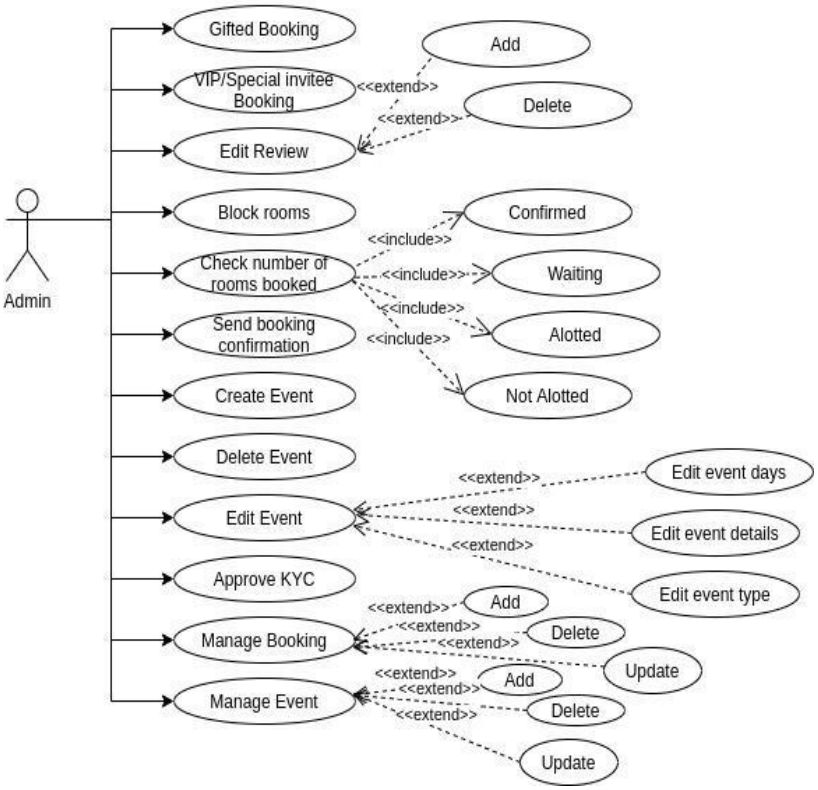


Fig 5.1.2 : Use Case-Advance



Fig 5.1.3 : Use Case Specifications

Use case description: The user inputs the data , KYC documents and booking functions to the system through a web interface. The system pre-processes the required data and KYC documents. Pre-processing phase contains activities like stop words removal, word stemming and document verifying. The system contains data w.r.t user with its unique credentials which classifies the user functionalities as desired.

5.2 Activity Diagram

Activity diagram description: The Activity diagram is presented for each main function in the project through Khoji(user) , admin and system end.It explains the detailed processing for each function and what happens from each end taken from inserting data ,updating , deleting and updating the same.

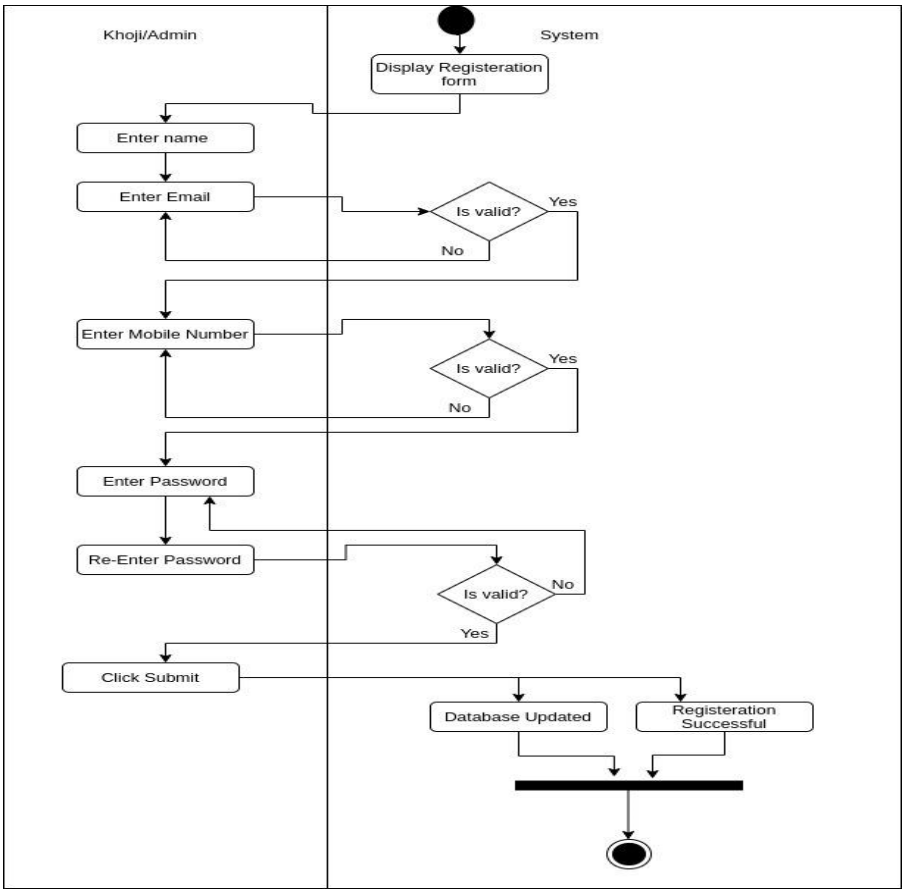


Fig 5.2.1: Activity diagram-Register

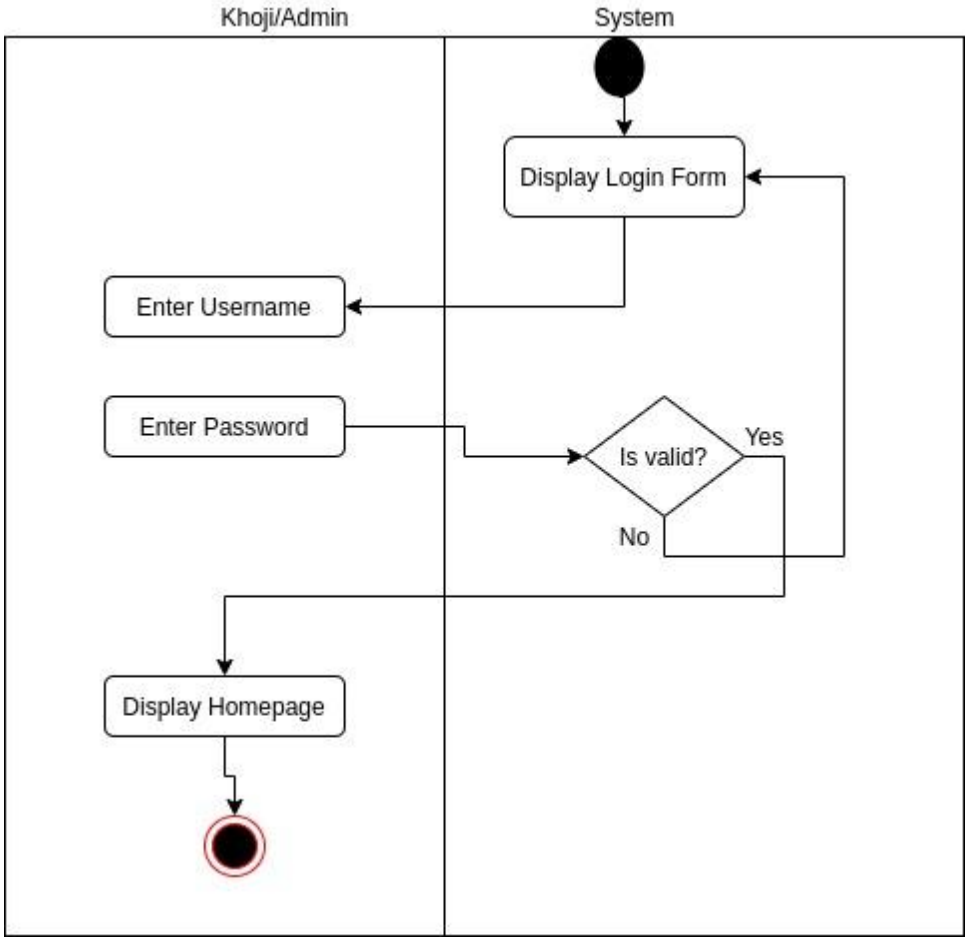


Fig 5.2.2: Activity diagram-Login

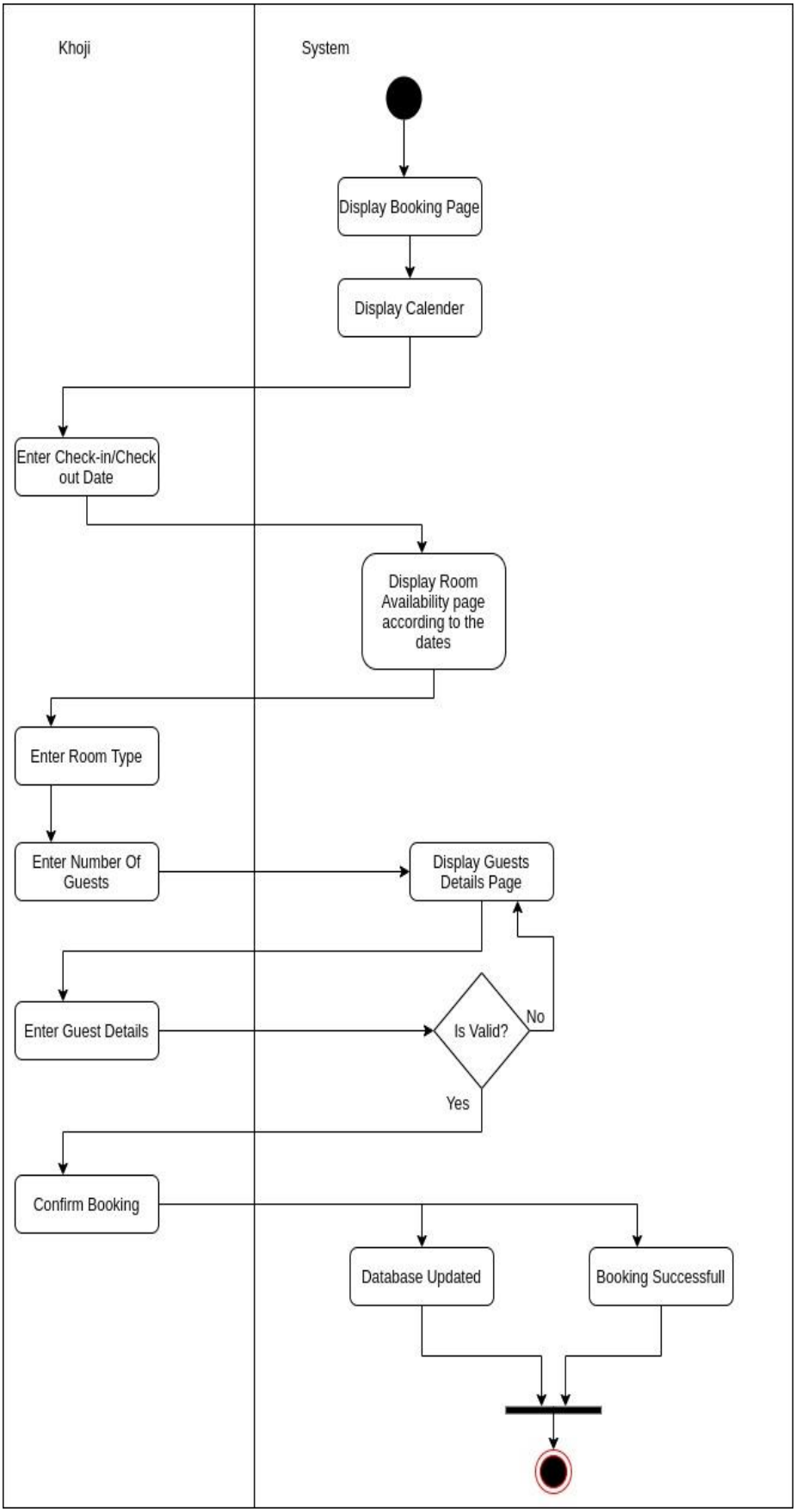


Fig 5.2.3: Activity Diagram-Add Booking

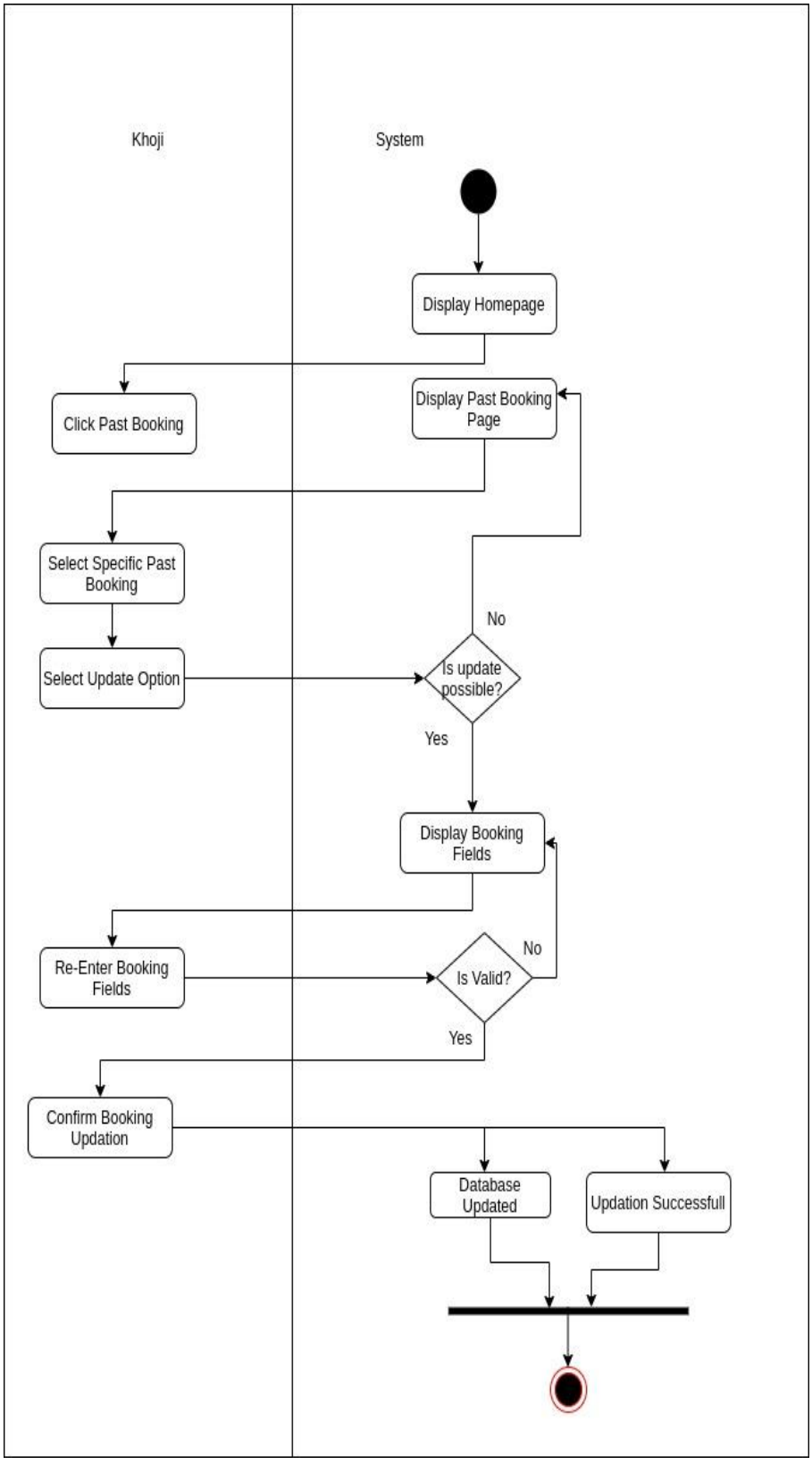


Fig 5.2.4: Activity Diagram-Update Booking

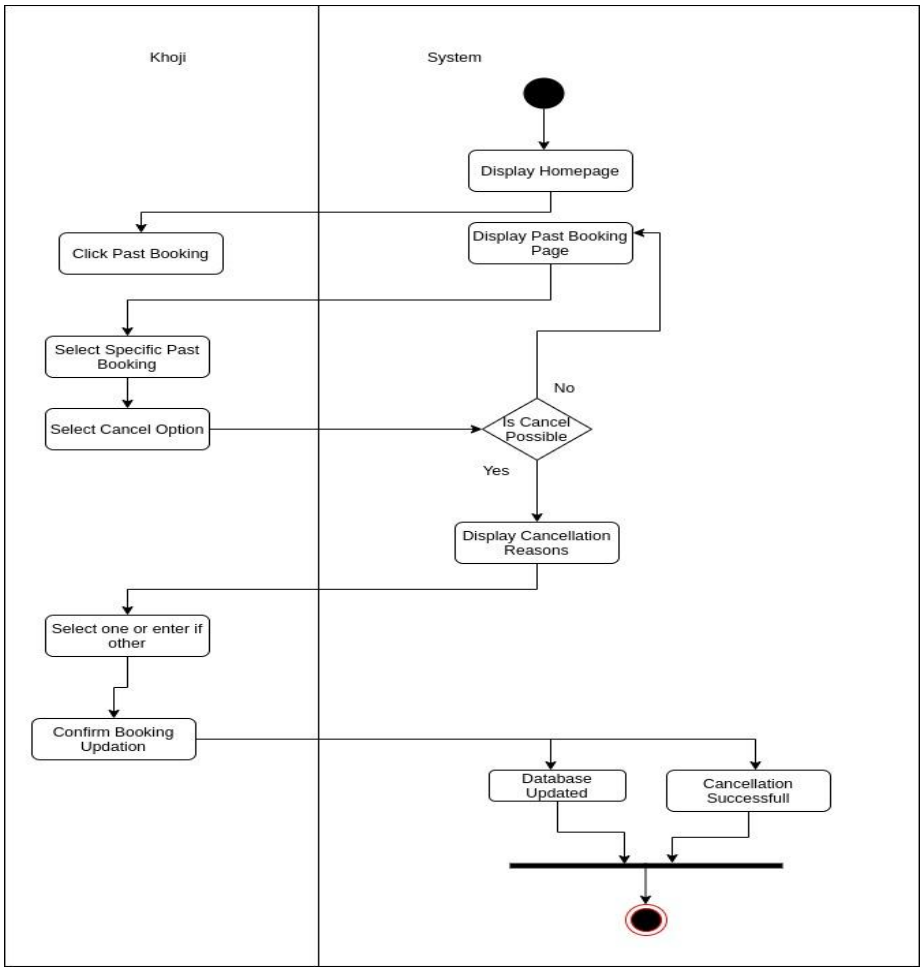


Fig 5.2.5: Activity Diagram-Cancel Booking

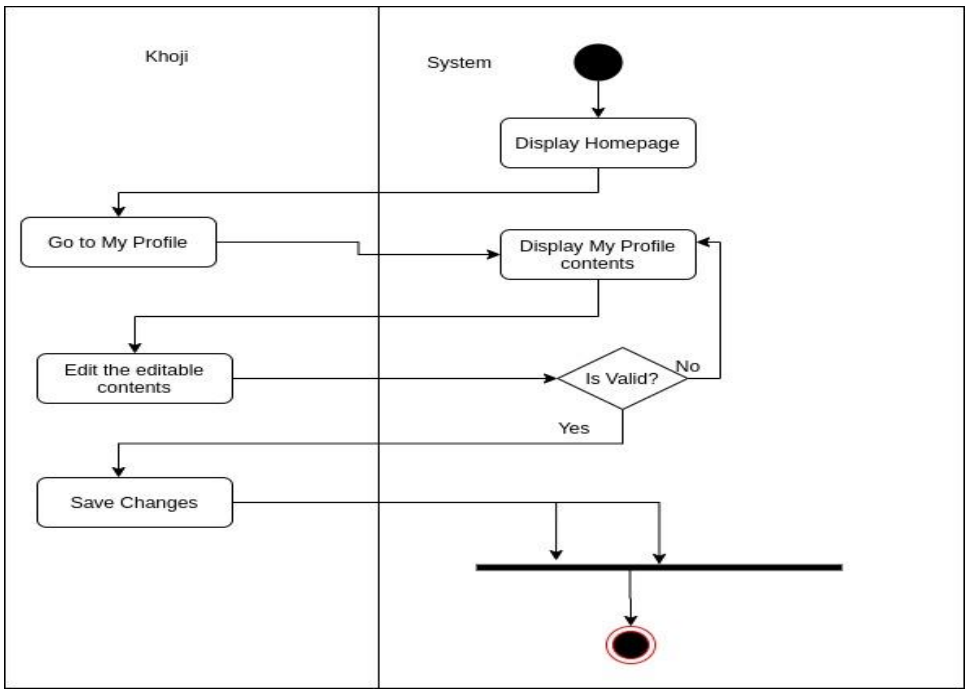


Fig 5.2.6: Edit Profile

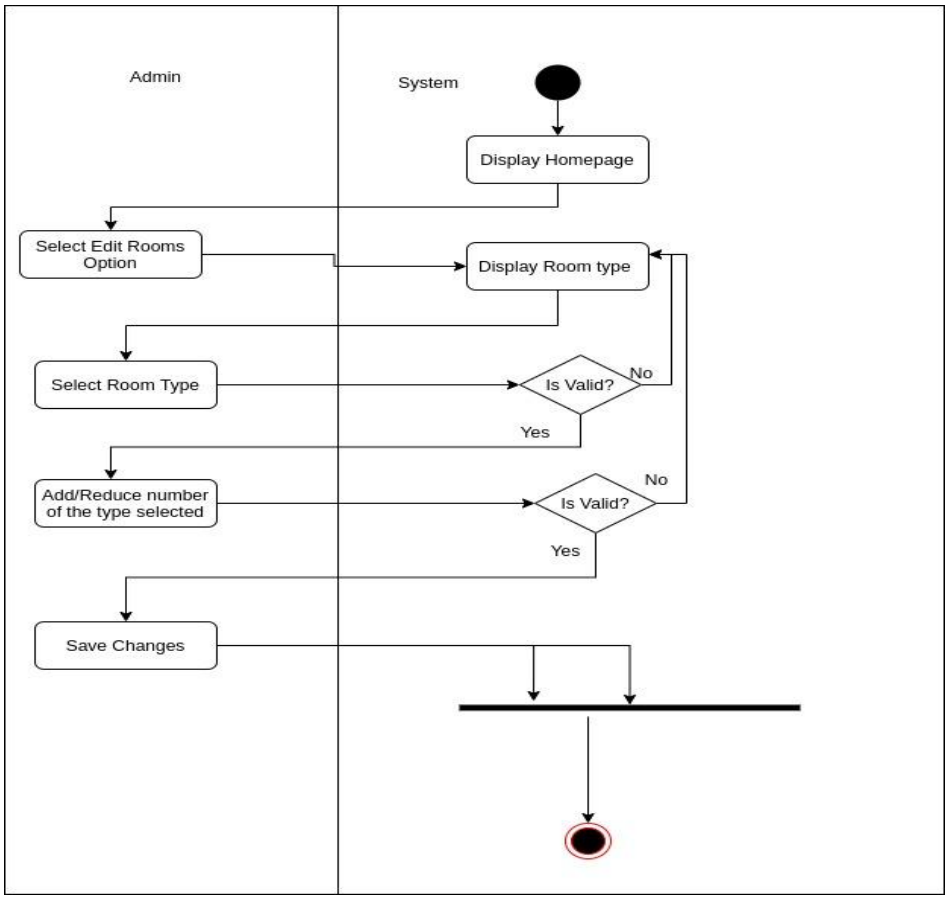


Fig 5.2.7: Edit Rooms

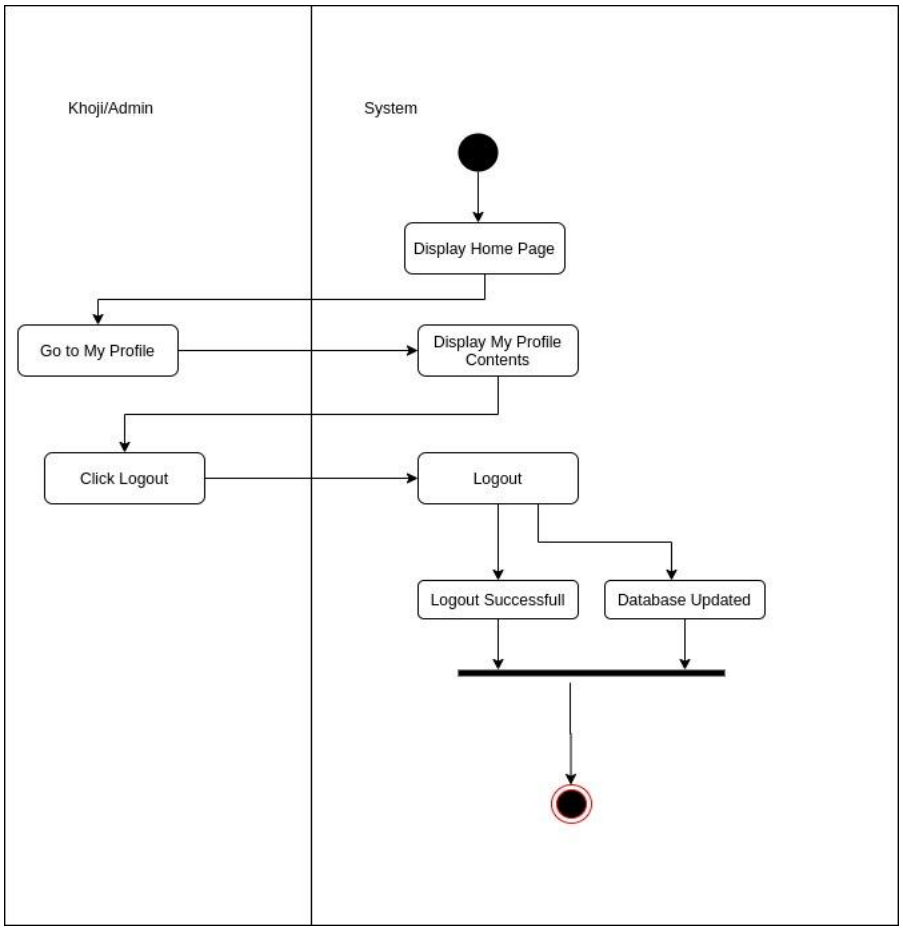


Fig 5.2.7: Logout

5.3 Sequence Diagram

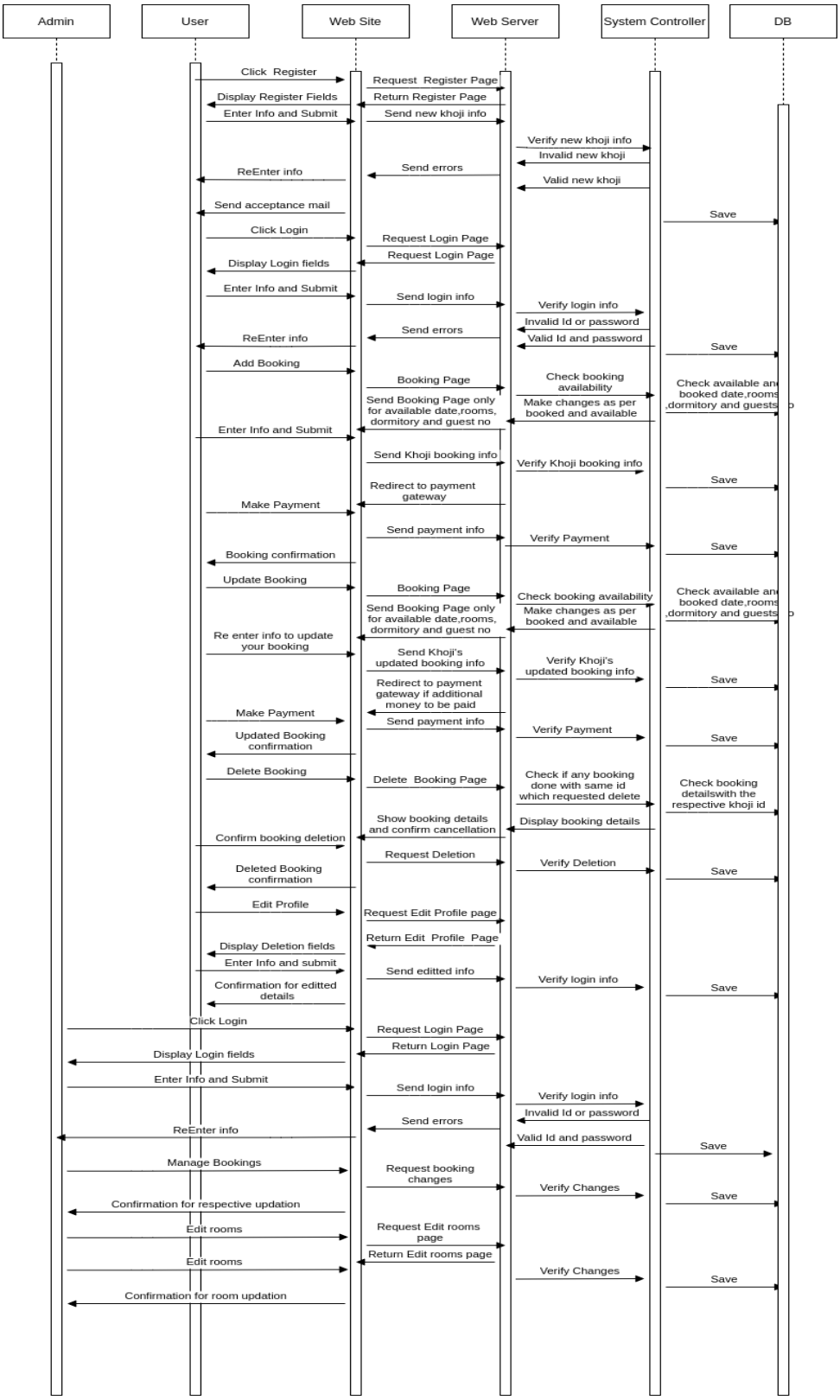
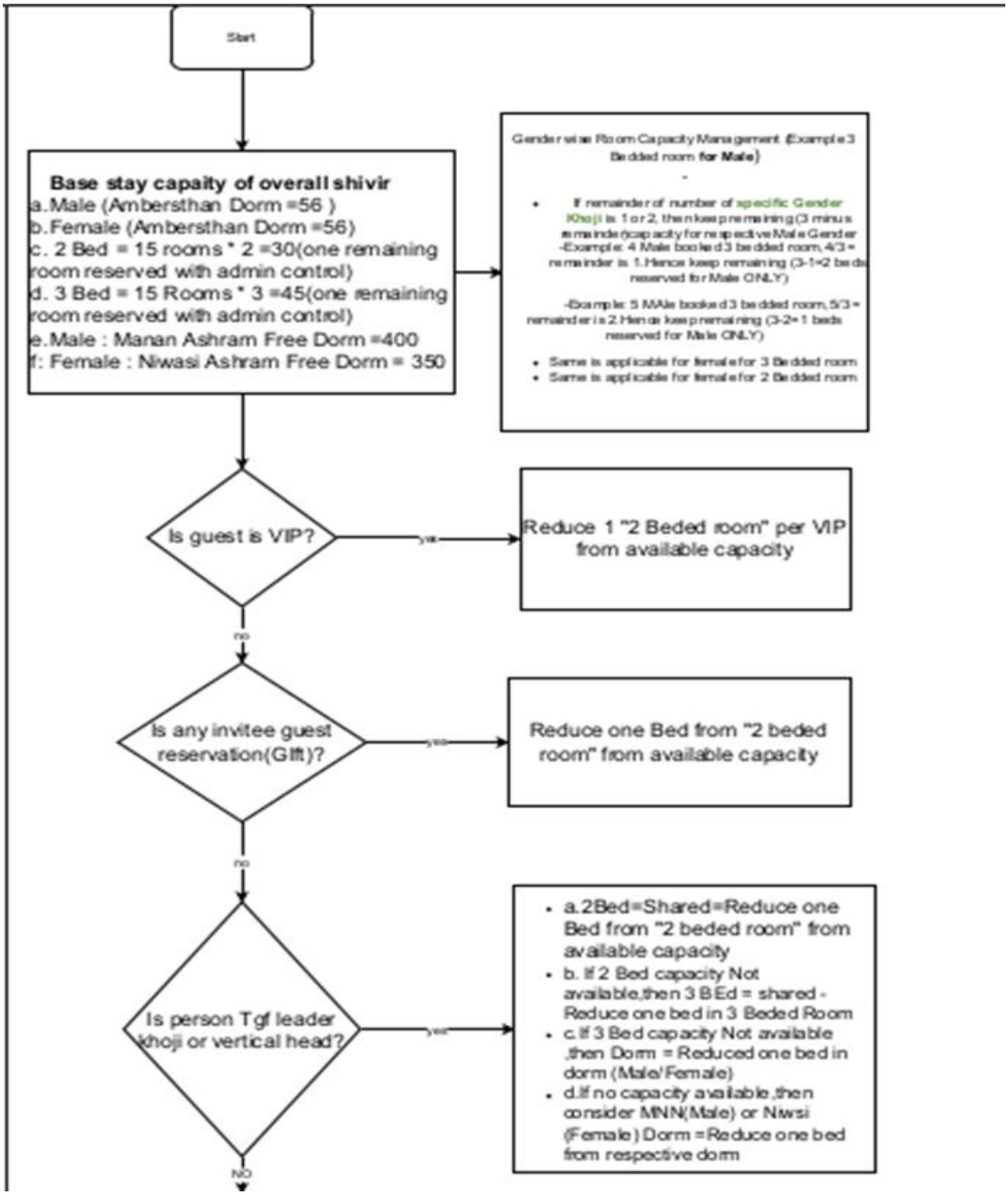


Fig 5.3 : Sequence Diagram

Sequence Diagram description: The user first communicates with the system through interface enters necessary data. System checks the data, verifies and sends it to the db. The system contains format ,necessary data and unique credentials of users which classifies the bookings then according to its functions and expected outcomes the data is processed , updated and saved therefore giving the necessary confirmation.

5.4 Workflow

- Capacity Management – For Capacity Management if the user for whom the accommodation is being booked is a VIP/special invitee then a whole room is booked and shown allotted in database. If the user is a normal Khoji or TGF leader then a bed from the particular room is being booked and shown allotted.



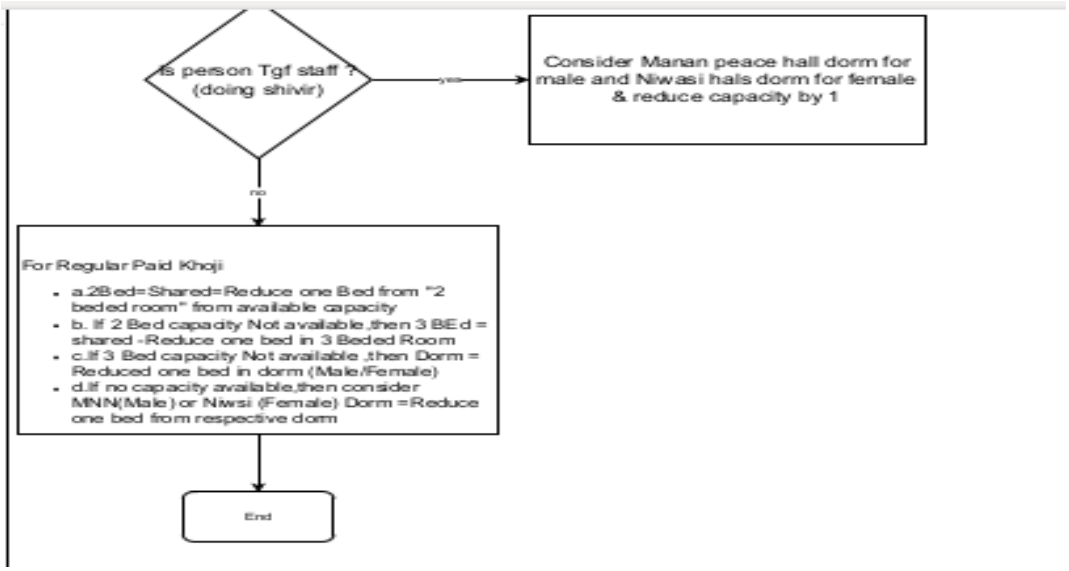


Fig 6.2.1 Capacity Management

- Gifted Booking – In case of gifted booking , the payment part of the respective Khoji is being waived off by the admin.

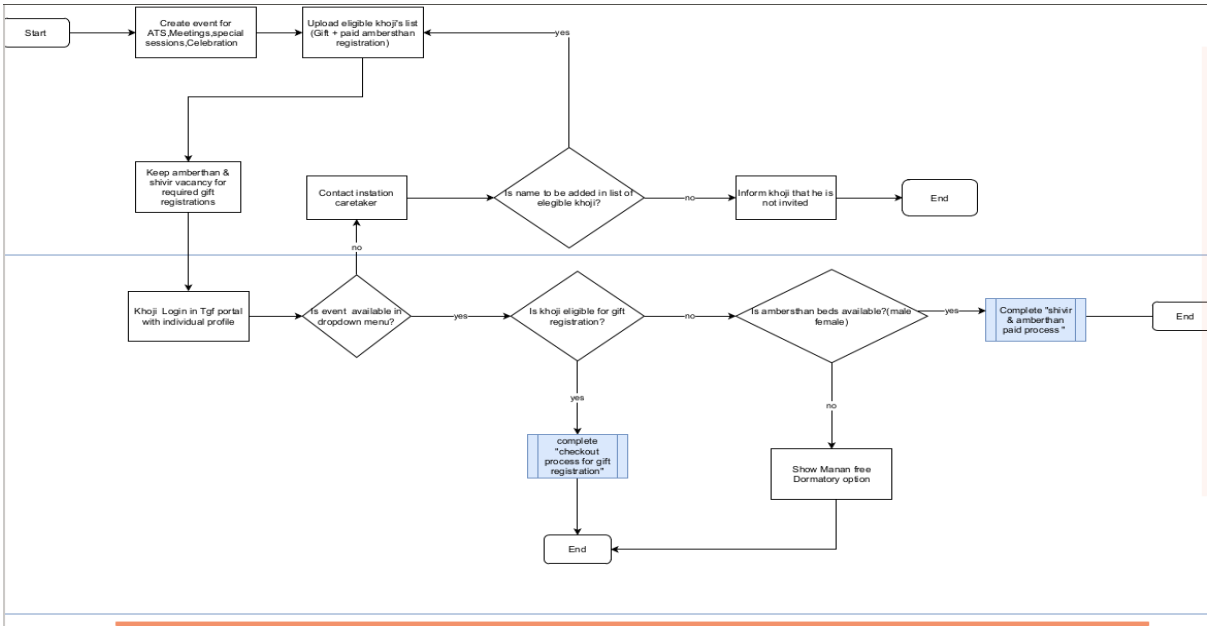


Fig 6.2.2 Gifted Booking

- Registeration – To complete this process, the Khoji first needs to select the dates in the calendar if shivir is available then the Khoji is directed to the available accommodation from which Khoji selects the types of accommodation based on availability and of guests which is further directed towards the kyc and payment.

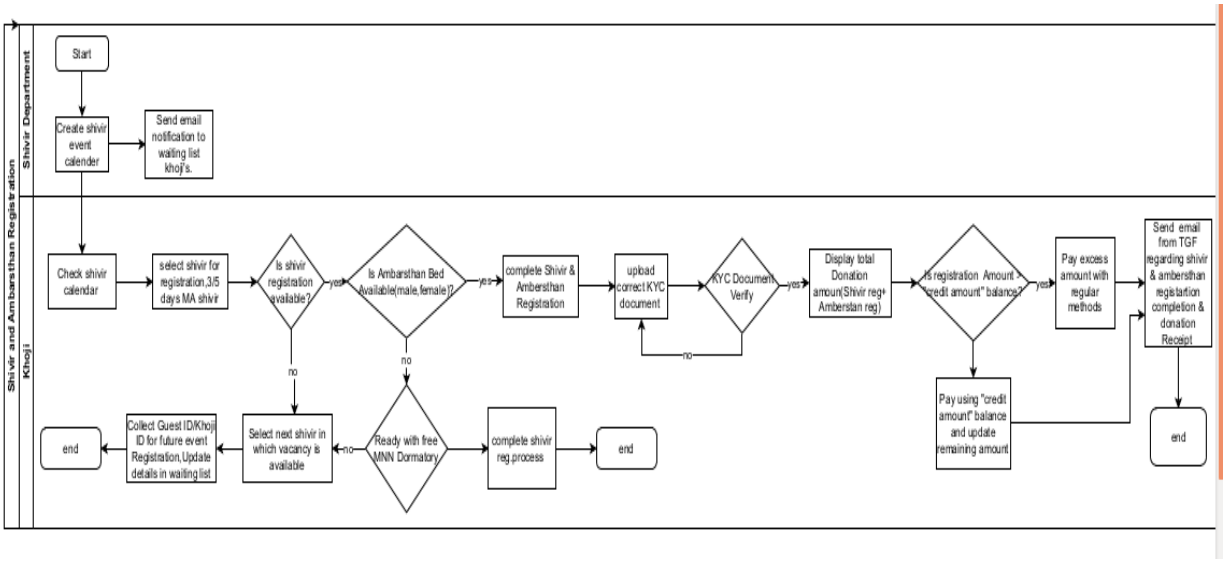


Fig 6.2.3 Registeration

- Non – Shivir days registeration – In this case khiji can book accommodation in case there is no event/shivir scheduled for the respective dates and can enjoy the benefit of staying and enjoying the accommodation on non shivir days

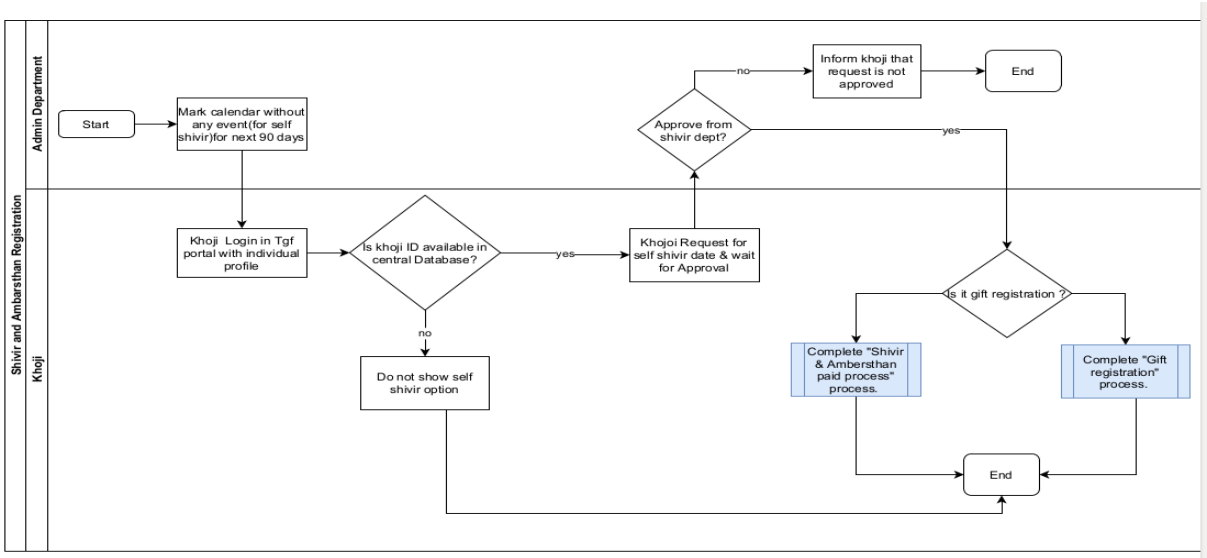


Fig 6.2.4 Non Shivir Days Registeration

5.5Time Schedule

Table No.5.5: Time Schedule

Task	Start	End	Status
Formation of problem statement ,scope, objective, business objective of the project and learning Phase	June	July	Completed
Formation of the system architecture and Brainstorming	July	September	Completed
Wireframes, functionalities and Api's	September	October	Completed
Developing UI screens , backend api services and integration.	October	November	Completed
Making the final report, collecting feedback from project guide ,knowledge transfer	November	December	Completed

5.5 Team Organization

5.5.1 Team Structure

Our team :- Our team consists of developers , internal guide , external guide and few TCS mentors.

Developers :

- Ms. Anisha Gandhi
- Mr. Pratik Jagtap
- Ms. Saloni Bonde
- Mr. Vedant Bhatambrekar

Internal Guide :

- Mrs.Suruchi Dedgaonkar

External Guide

- Mr. Rajendra Rode
- Mr. Deepak Purnaye
- Mr. Ishwar Gade

TCS mentors

- Mr. Sham Phadtale
- Ms/Mrs. Sarika Nesarikar
- Ms/Mrs. Aarti
- Mr. Aneek

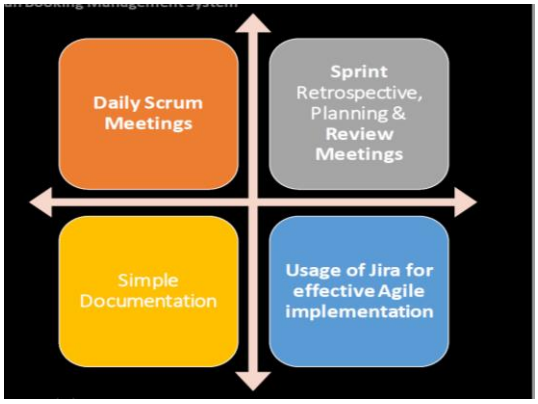
5.5.2 Task Summary

Member	Tasks
Common	Discussion with industry on Choice of Technologies, Software Development methodology, Scope Finalization, Legal Consultations, Business Logic Development, Modules Formation, Sprint-wise Documentation.
Saloni Bonde	Frontend Wireframes on Framer(All the modules), Worked on the Screens on Sign-up for three types of guests with angular, Sign-in & Reset Password. Integration of Frontend with backend.
Vedant Bhatambrekar	Frontend Wireframes on Framer(All the modules), Worked on the Screens on Sign-up for three types of guests with angular, Sign-in & Reset Password, Integration of Frontend with backend.
Pratik Jagtap	Database designing for centralized team & Ambarsthan, APIs in Springboot, Integration of Frontend with backend.
Anisha Gandhi	Database designing for centralized team & Ambarsthan, APIs in Springboot, Object Oriented Diagrams.

5.5.2 Agile Methodology Practice

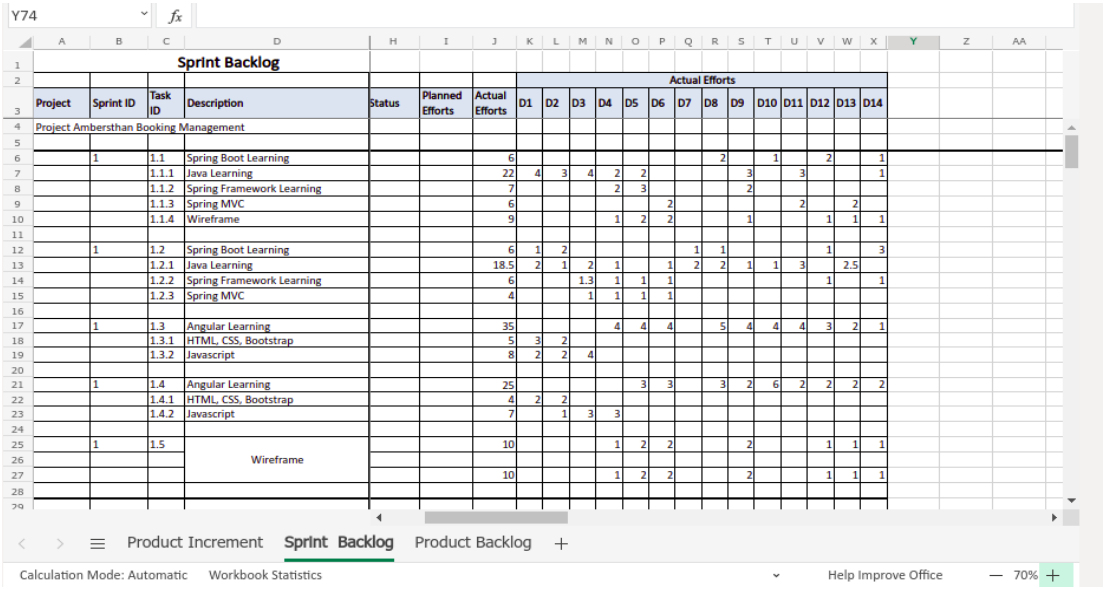
In our project, we followed agile methodology. AGILE methodology is a practice that promotes continuous iteration of development and testing throughout the software development life cycle of the project. In the Agile model, both development and testing activities are concurrent.

- Daily scrum calls : In our daily scrum calls (Duration 15-30 minutes) in the presence of VIIT faculty Mrs. Suruchi Dedgaonkar , TGF representative Mr. Rajendra Rode and all four project group members we discussed about the daily updates of work , roadblocks and progress regarding the same. Also , we planned the task heading up.



- Weekly Review Meeting : In our weekly review meeting we discussed the progress of work done in that particular week , the work that should be done in the week showing up next.Also , we discussed all possible progress and roadblock factors and solutions towards the same.

- Scrum Artefacts : We portrayed an excel sheet on a sprint basis (1 Sprint = 2 weeks) which describes tasks to be done , by whom to be done and number of hours worked towards the same.



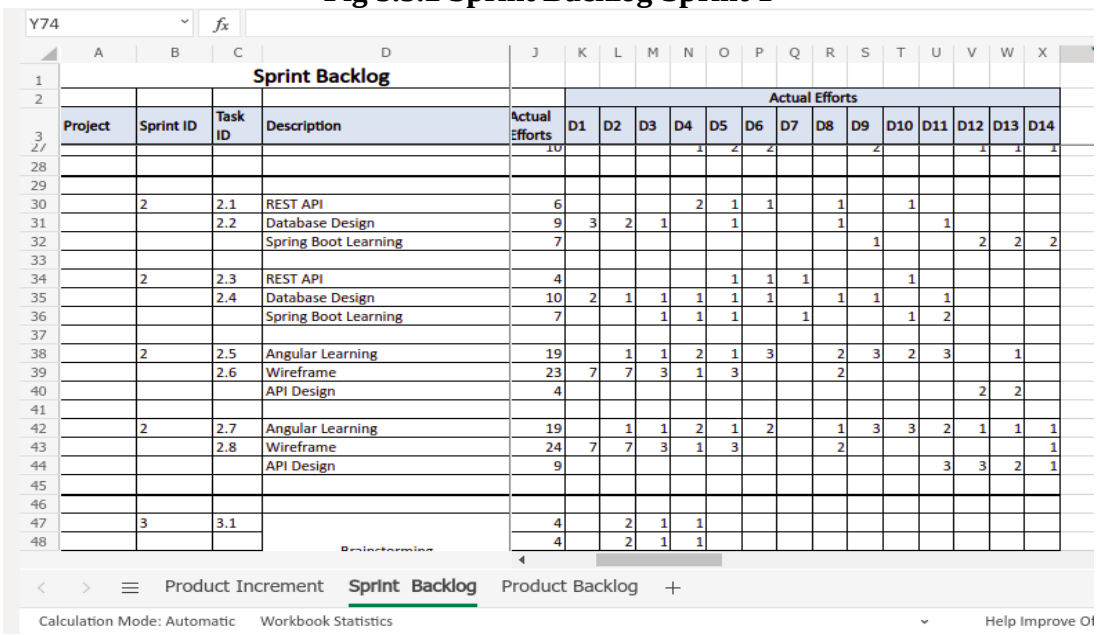
Sprint Backlog

Project	Sprint ID	Task ID	Description	Status	Planned Efforts	Actual Efforts	Actual Efforts D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
Project Amberstha Booking Management																				
	1	1.1	Spring Boot Learning			6								2		1		2		1
		1.1.1	Java Learning			22	4	3	4	2	2				3		3			1
		1.1.2	Spring Framework Learning			7				2	3				2					
		1.1.3	Spring MVC			6						2					2		2	
		1.1.4	Wireframe			9				1	2	2			1			1	1	1
	1	1.2	Spring Boot Learning			6	1	2					1	1				1		3
		1.2.1	Java Learning			18.5	2	1	2	1		1	2	2	1	1	3		2.5	
		1.2.2	Spring Framework Learning			6			1.3	1	1	1						1		1
		1.2.3	Spring MVC			4			1	1	1	1								
	1	1.3	Angular Learning			35				4	4	4		5	4	4	4	3	2	1
		1.3.1	HTML, CSS, Bootstrap			5	3	2												
		1.3.2	Javascript			8	2	2	4											
	1	1.4	Angular Learning			25					3	3		3	2	6	2	2	2	2
		1.4.1	HTML, CSS, Bootstrap			4	2	2												
		1.4.2	Javascript			7		1	3	3										
	1	1.5	Wireframe			10				1	2	2			2			1	1	1
						10				1	2	2			2			1	1	1

Product Increment | Sprint Backlog | Product Backlog

Calculation Mode: Automatic Workbook Statistics Help Improve Office 70%

Fig 5.5.2 Sprint Backlog Sprint 1



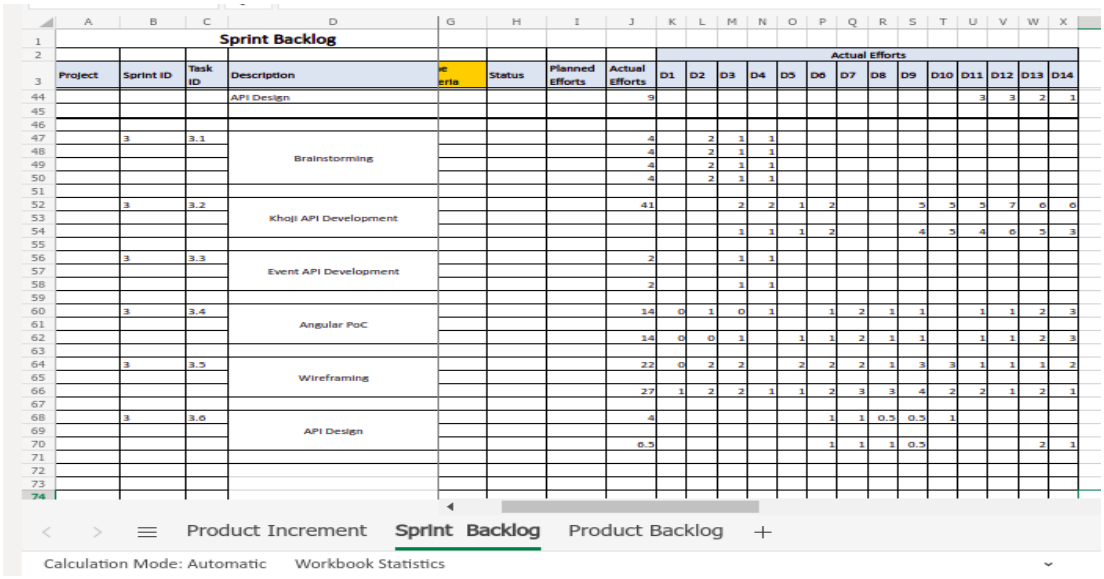
Sprint Backlog

Project	Sprint ID	Task ID	Description	Actual Efforts	Actual Efforts D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
				10				1	2	2			2			1	1	1
	2	2.1	REST API	6				2	1	1		1		1				
		2.2	Database Design	9	3	2	1		1			1			1			
			Spring Boot Learning	7									1			2	2	2
	2	2.3	REST API	4					1	1	1			1				
		2.4	Database Design	10	2	1	1	1	1	1		1	1		1			
			Spring Boot Learning	7			1	1	1		1			1	2			
	2	2.5	Angular Learning	19		1	1	2	1	3		2	3	2	3		1	
		2.6	Wireframe	23	7	7	3	1	3			2						
			API Design	4												2	2	
	2	2.7	Angular Learning	19		1	1	2	1	2		1	3	3	2	1	1	1
		2.8	Wireframe	24	7	7	3	1	3			2						1
			API Design	9											3	3	2	1
	3	3.1	Brainstorming	4		2	1	1										
				4		2	1	1										

Product Increment | Sprint Backlog | Product Backlog

Calculation Mode: Automatic Workbook Statistics Help Improve Of

Fig 5.5.2 Sprint Backlog Sprint 2



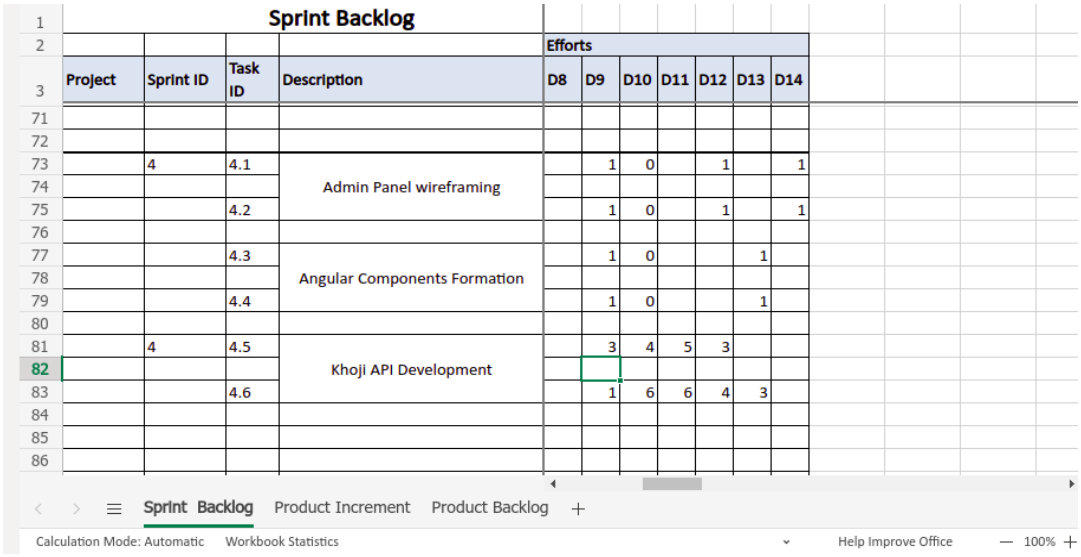
Sprint Backlog

Project	Sprint ID	Task ID	Description	Status	Planned Efforts	Actual Efforts	Actual Efforts D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
			API Design			9											3	3	2	1
	3	3.1	Brainstorming			4		2	1	1										
						4		2	1	1										
						4		2	1	1										
						4		2	1	1										
	3	3.2	Khoji API Development			41			2	2	1	2			5	5	5	7	6	6
									1	1	1	2			4	5	4	6	5	3
	3	3.3	Event API Development			2			1	1										
						2			1	1										
	3	3.4	Angular PoC			14	0	1	0	1		1	2	1	1		1	1	2	3
						14	0	0	1		1	1	2	1	1		1	1	2	3
	3	3.5	Wireframing			22	0	2	2		2	2	2	1	3	3	1	1	1	2
						27	1	2	2	1	1	2	3	3	4	2	2	1	2	1
	3	3.6	API Design			4						1	1	0.5	0.5	1				
						6.5						1	1	1	0.5				2	1

Product Increment | Sprint Backlog | Product Backlog

Calculation Mode: Automatic Workbook Statistics

Fig 5.5.2 Sprint Backlog Sprint 3



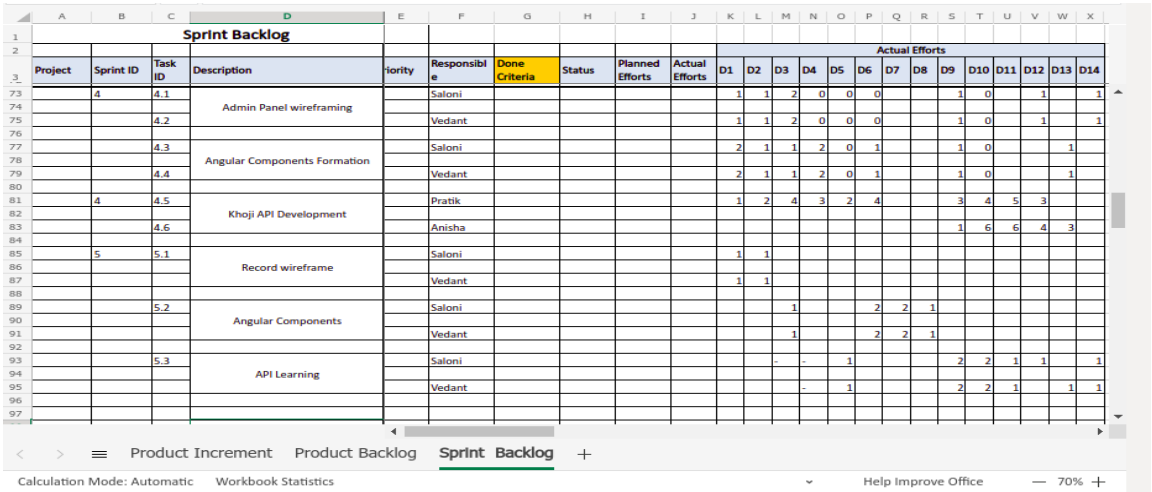
Sprint Backlog

Project	Sprint ID	Task ID	Description	Efforts D8	D9	D10	D11	D12	D13	D14
	4	4.1	Admin Panel wireframing		1	0		1		1
		4.2			1	0		1		1
		4.3	Angular Components Formation		1	0			1	
		4.4			1	0			1	
	4	4.5	Khoji API Development		3	4	5	3		
		4.6			1	6	6	4	3	

Sprint Backlog | Product Increment | Product Backlog

Calculation Mode: Automatic Workbook Statistics Help Improve Office 100%

Fig 5.5.2 Sprint Backlog Sprint 4



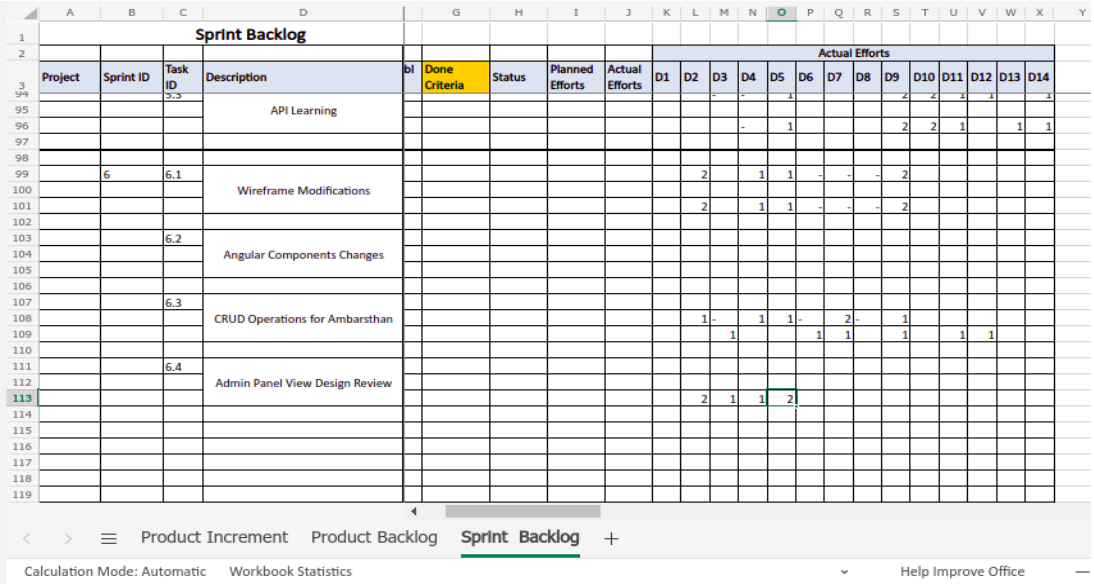
Sprint Backlog

Project	Sprint ID	Task ID	Description	Priority	Responsible	Done Criteria	Status	Planned Efforts	Actual Efforts	Actual Efforts D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
	4	4.1	Admin Panel wireframing		Saloni					1	1	2	0	0	0			1	0		1		1
		4.2			Vedant					1	1	2	0	0	0			1	0		1		1
		4.3	Angular Components Formation		Saloni					2	1	1	2	0	1			1	0			1	
		4.4			Vedant					2	1	1	2	0	1			1	0			1	
	4	4.5	Khoji API Development		Pratik					1	2	4	3	2	4			3	4	5	3		
		4.6			Anisha													1	6	6	4	3	
	5	5.1	Record wireframe		Saloni					1	1												
					Vedant					1	1												
		5.2	Angular Components		Saloni							1			2	2	1						
					Vedant							1			2	2	1						
		5.3	API Learning		Saloni							-	-	1				2	2	1	1		1
					Vedant								-	1				2	2	1		1	1

Product Increment | Product Backlog | Sprint Backlog

Calculation Mode: Automatic Workbook Statistics Help Improve Office 70%

Fig 5.5.2 Sprint Backlog Sprint 5



Sprint Backlog

Project	Sprint ID	Task ID	Description	bl	Done Criteria	Status	Planned Efforts	Actual Efforts	Actual Efforts D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14
			API Learning									-	1				2	2	1			
												-	1				2	2	1		1	1
	6	6.1	Wireframe Modifications							2		1	1	-	-	-	2					
										2		1	1	-	-	-	2					
		6.2	Angular Components Changes																			
		6.3	CRUD Operations for Ambarsthan							1	-	1	1	-	2	-	1					
											1			1	1		1		1	1		
		6.4	Admin Panel View Design Review																			
										2	1	1	2									

Product Increment | Product Backlog | Sprint Backlog

Calculation Mode: Automatic | Workbook Statistics | Help Improve Office

Fig 5.5.3 Sprint Backlog Sprint 6

Sprint #	Start Date	End Date	Sprint Scope	Remarks
1	13-06-20	26-06-20	1.Springboot Learning Angular Learning Wireframe	Wireframe using Invision was not appealing
2	26-06-20	11-07-20	1.Database Design REST API Design Wireframe using Framer	-API related queries addressed by Sham Sir -Database designed and sent for approva -Wireframe modified
3	11-07-20	26-07-20	1.Khoji API Development Angular PoC Wireframe updation according to expert suggestions API design	-API development in process -Wireframe for administrator in process -Suggested task for Angular PoC and API design completed

Sprint #	Start Date	End Date	Sprint Scope	Remarks
4	27-07-20	8-08-20	1.Angular component formation Khoji API Development Admin control panel design Brainstorming	-Queries asked to TGF regarding process and parameter finalization -API related queries addressed by Sham Sir
5	9-08-20	22-08-20	1.Wireframe recording for front end Angular components for front end Wireframe for Admin view Database review and fine tuning Khoji API development – tasks added Brainstorming	- In progress Some Queries answered by Sharad Sir regarding process and parameter finalization API related queries addressed and database related tasks assigned by Sham Sir

Sprint #	Start Date	End Date	Sprint Scope	Remarks
6	24-08-20	6-09-20	1.Wireframe design for Admin Panel Angular components updation Wireframe presentation recording CRUD operation in DB Khoji Api to retrieve data on the basis of combination like first and last name, mobile number. Cross learning in between teams	– Backend admin panel wireframe detailed design shared for review and feedback. Waiting for Angular Components Reviews Waiting for DB review so as to start CRUD operations on ABMS DB
7	7-09-20	19-09-20	1.Updating CURD operation in centralized DB Updating DB in workbench Designing of screen using HTML/CSS Cross learning Students preparation for placements	– Got modified requirements and flow Need review on Backend admin panel wireframe CRUD Operation in ABMS DB depends on DB review UI/UX meeting planned on Friday for Wireframe

Fig 5.2.3 Sprint Snapshots

5.5.3 Tools : Project Management

- Microsoft Teams - For Daily / Weekly meetings , Extra learning sessions were held in teams , daily updates were posted through comments , buckets of the respective task were changed on basis of (to do , doing , done)

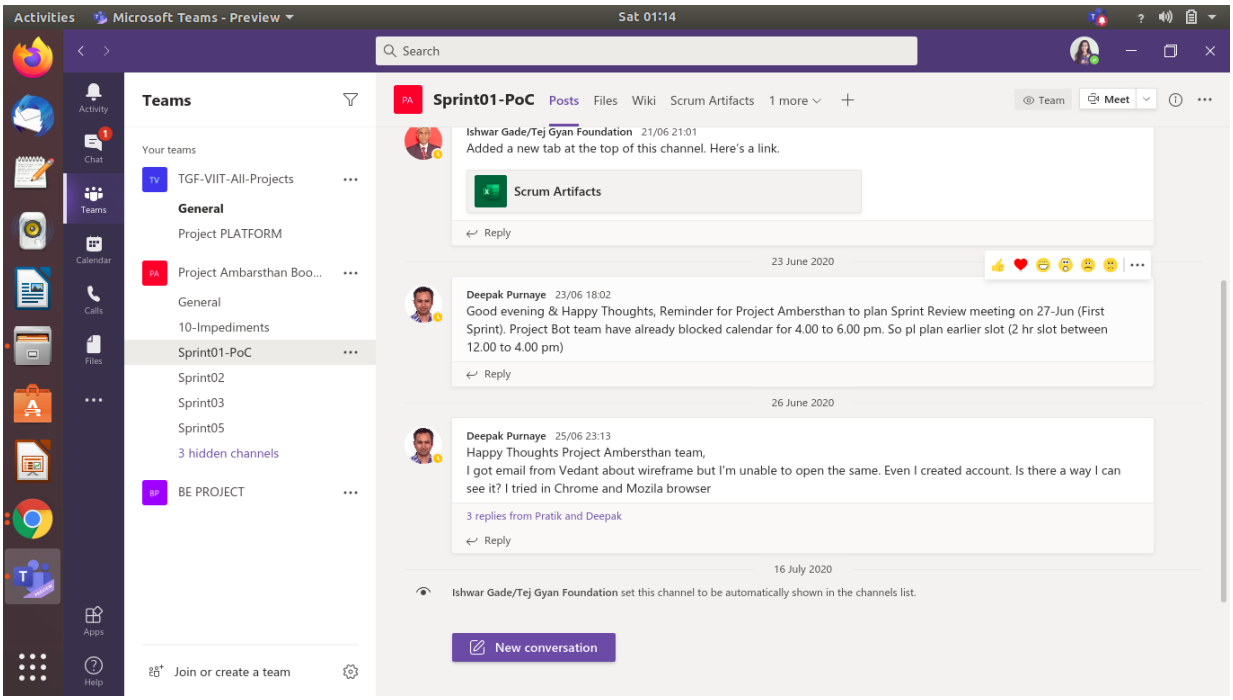


Fig 5.5.3 MS Teams

- Outlook – Conversation over Mails

- Git - Pushing the the actual implementation (code)

- Jira - User stories , daily comments on the respective user stories stating the updates.

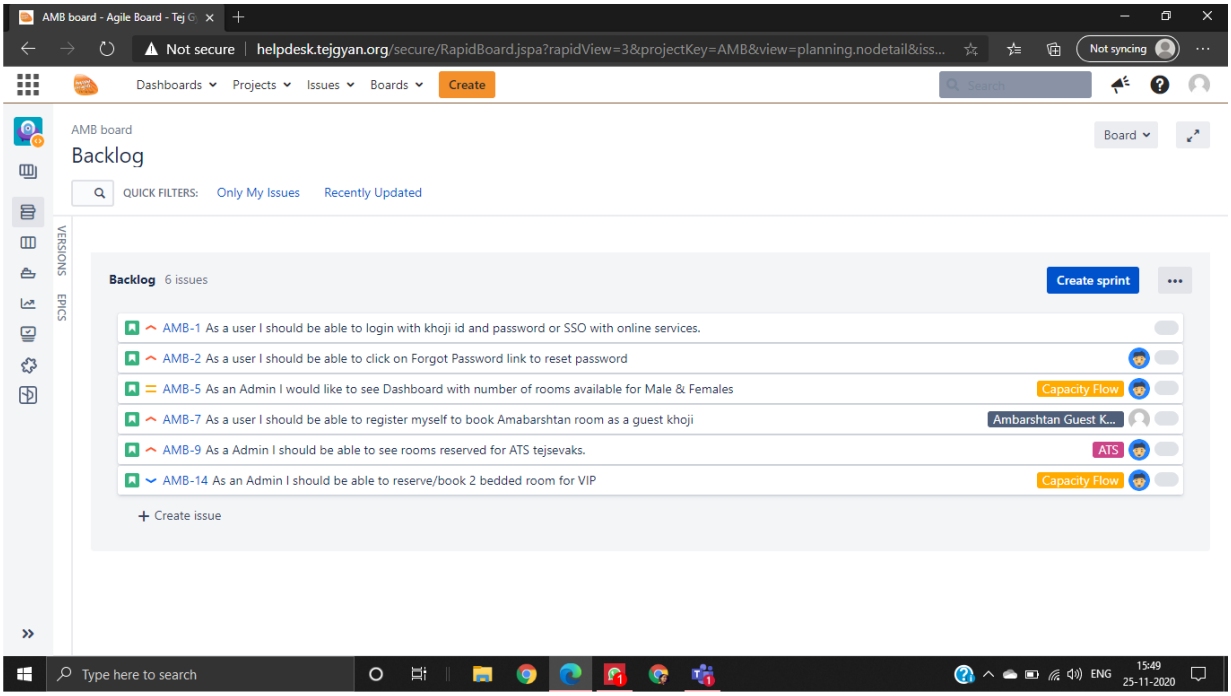


Fig : 5.5.3 Usage of Jira

5.5.4 Learning Sessions

Depending on the various learning phases in our project we had industry experts who conducted sessions on such topics.

Topics :

- Agile Methodology
- Angular
- Jira
- Springboot
- UI/UX
- JUnit

5.6 Interface Details and Screenshots

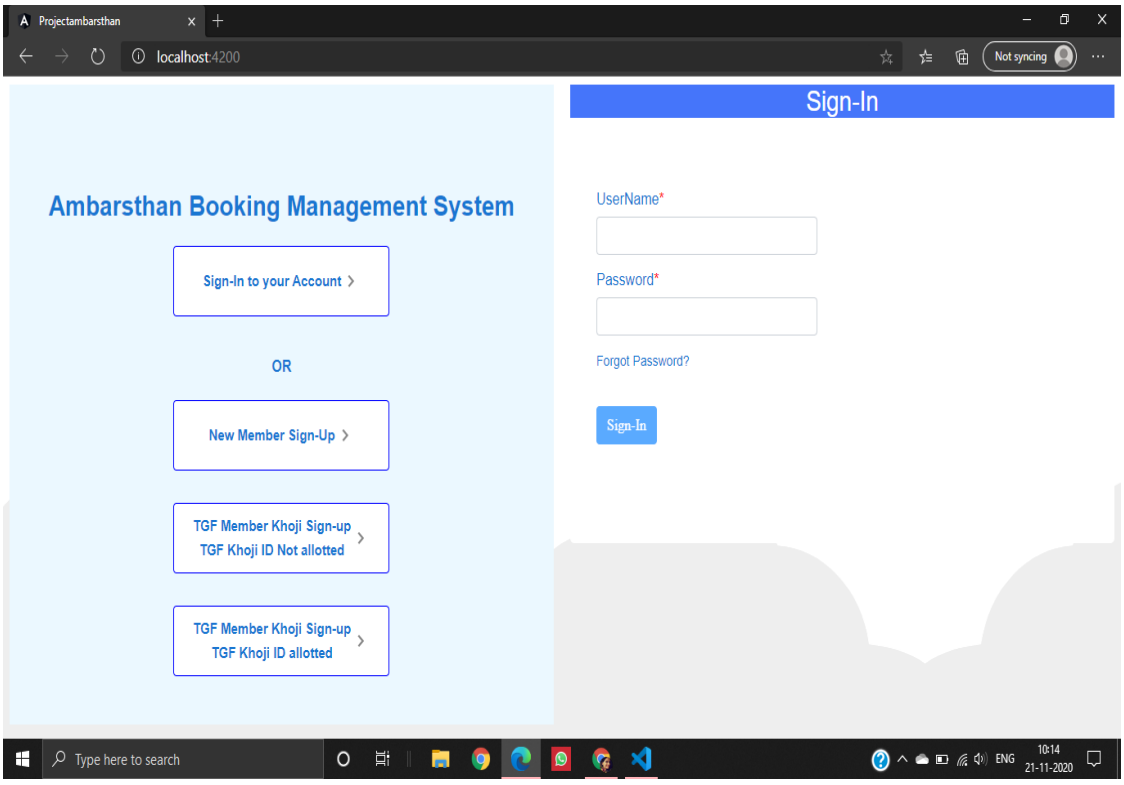


Fig 5.6.1 Sign-in

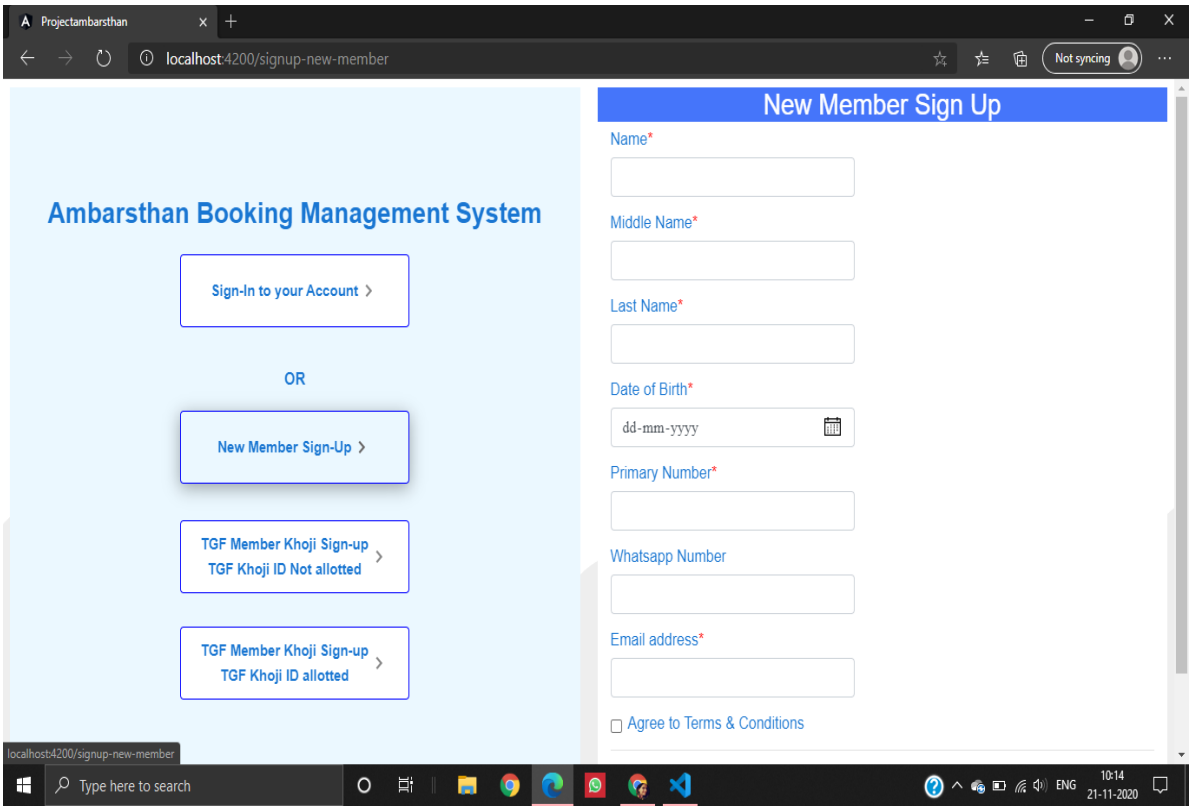


Fig 5.6.2 Sign-up

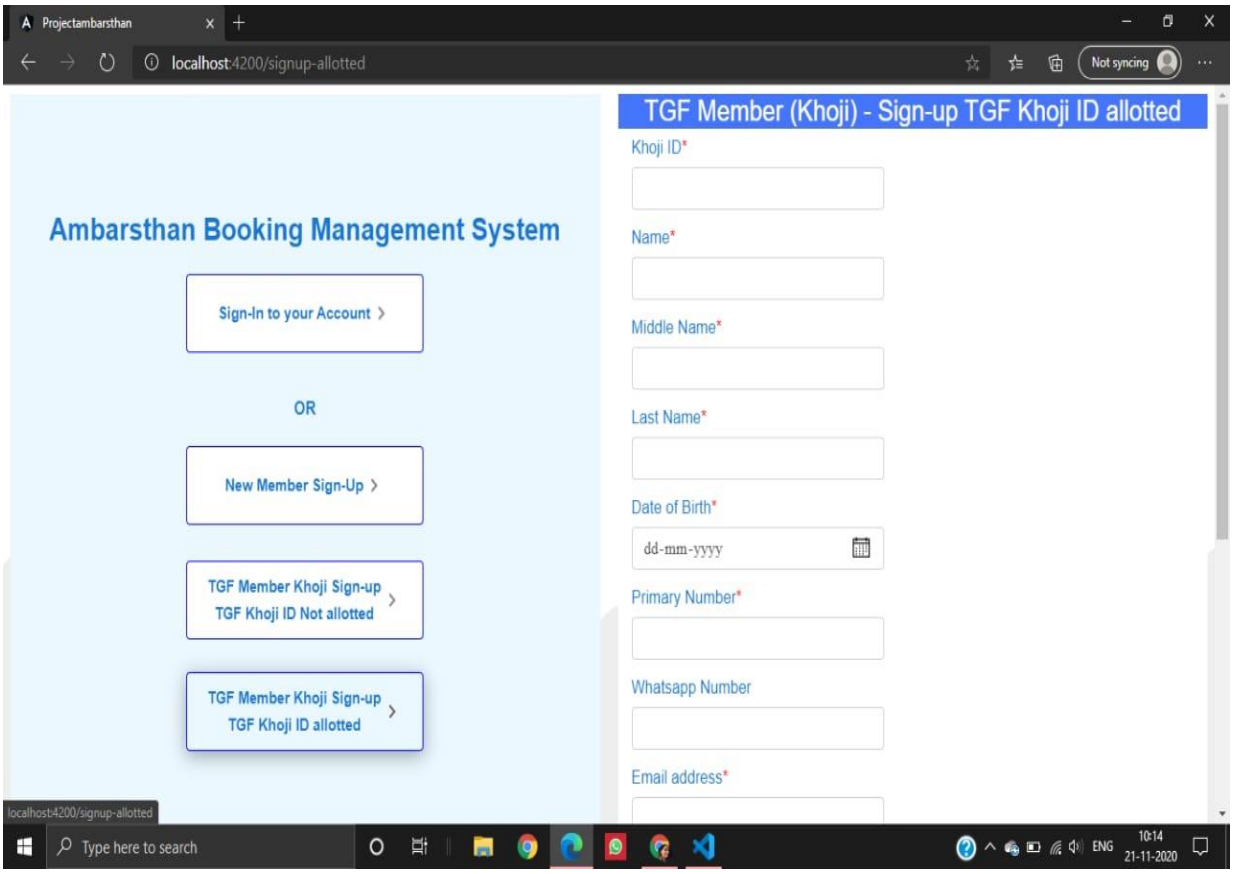


Fig 5.6.3 Sign-up (TGF Khoji ID allotted)

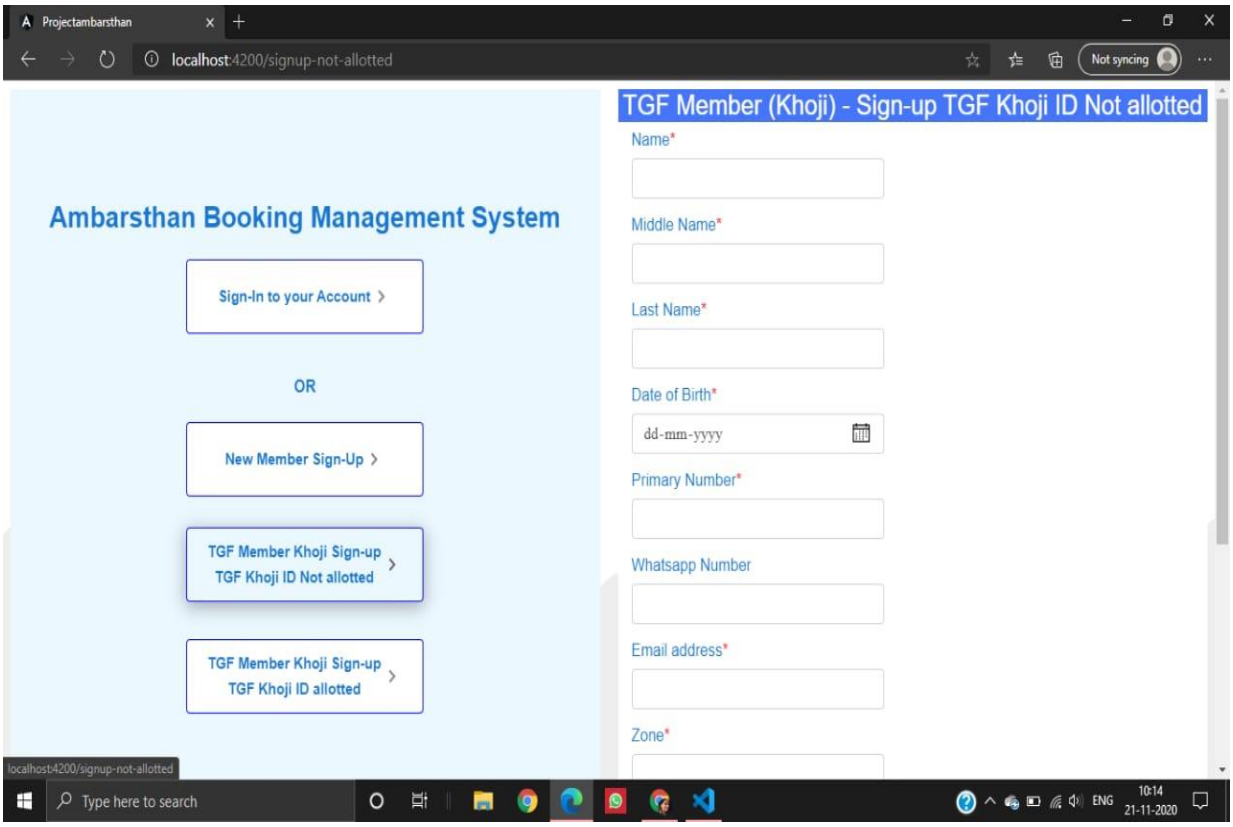


Fig 5.6.4 Sign-up(TGF Khoji ID not allotted)

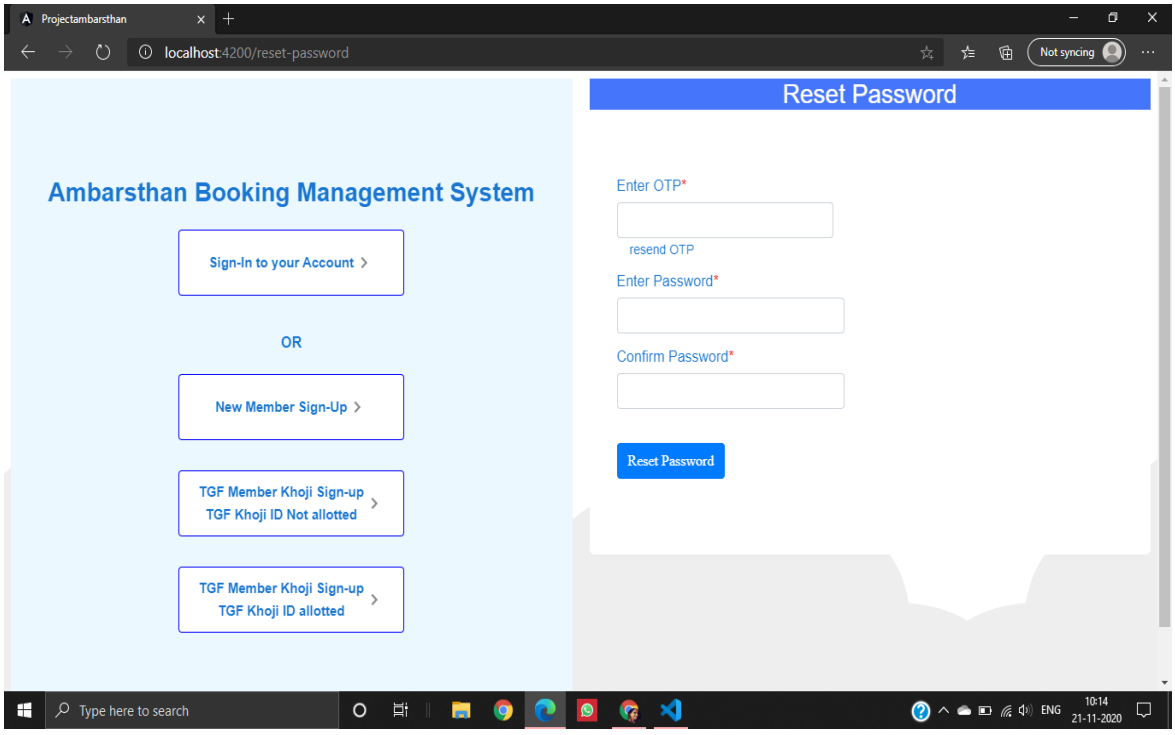


Fig 5.6.5 Reset Password

Chapter 6

Implementation Software Testing

6.1 Implementation



Fig

Fig.6.1.1 Wireframe – Homepage

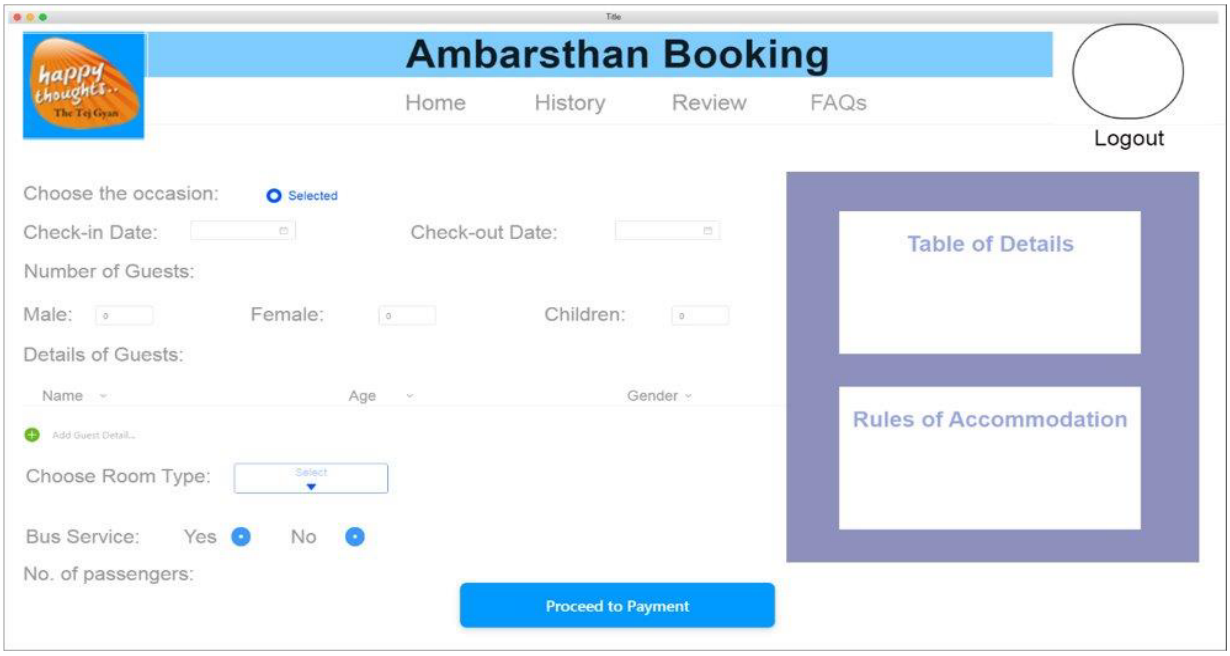


Fig 6.1.2 Wireframe – Booking Page

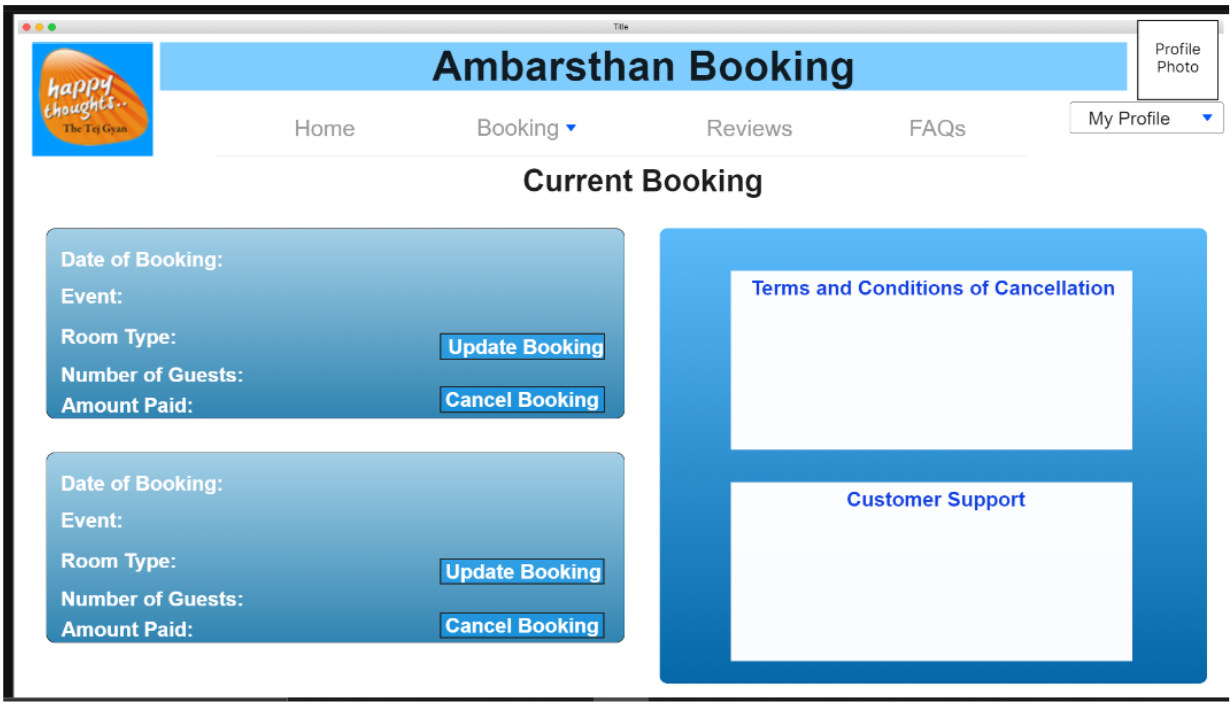


Fig 6.1.3 Wireframe – Update Booking Page

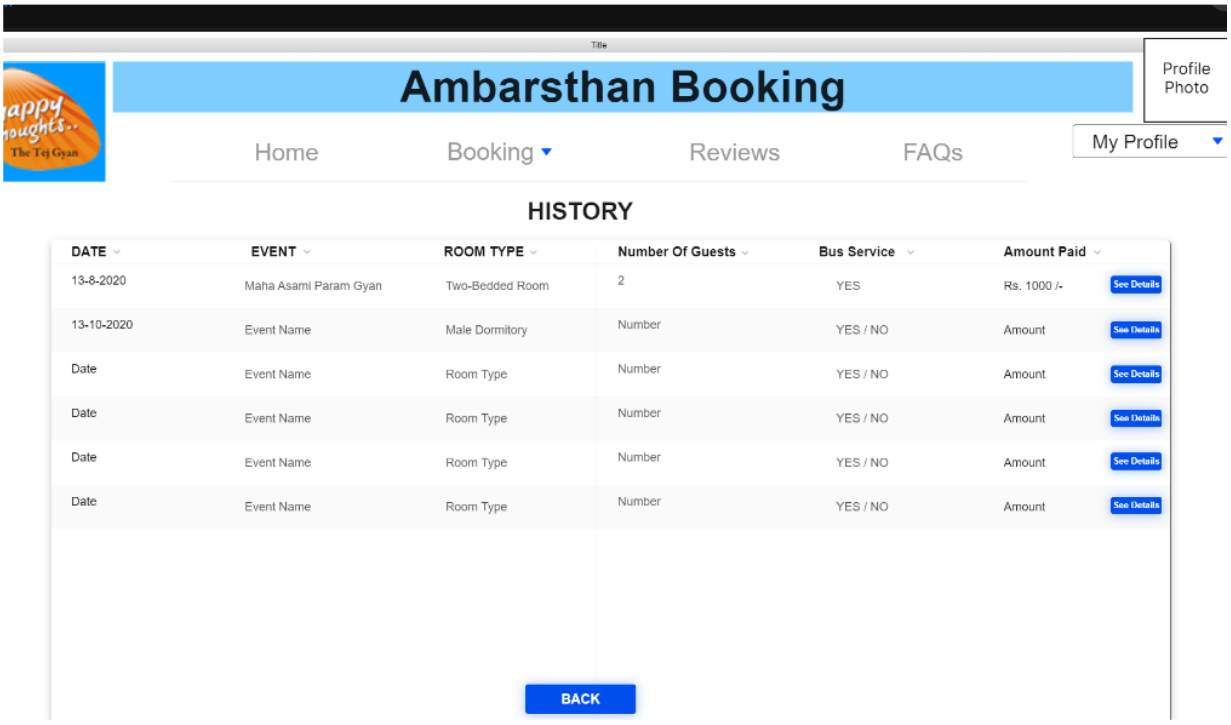


Fig 6.1.4 Wireframe – History

6.2 Database Creation

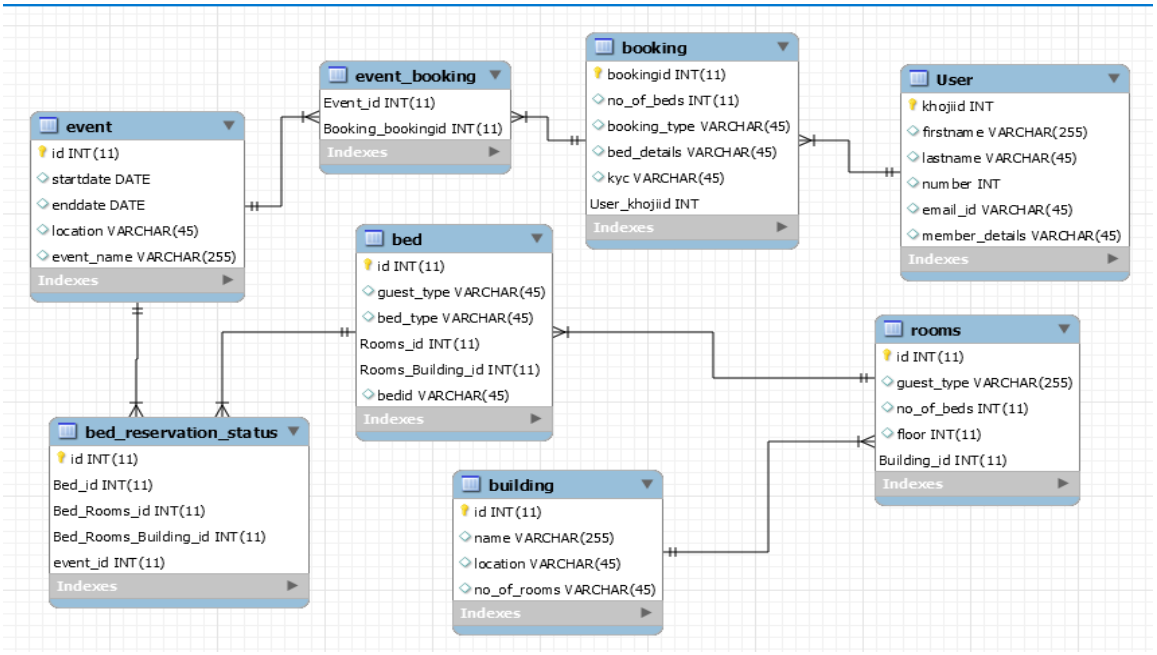


Fig 6.2. Database Design

6.3 Integration



Fig 6.3.1 API's Screenshot

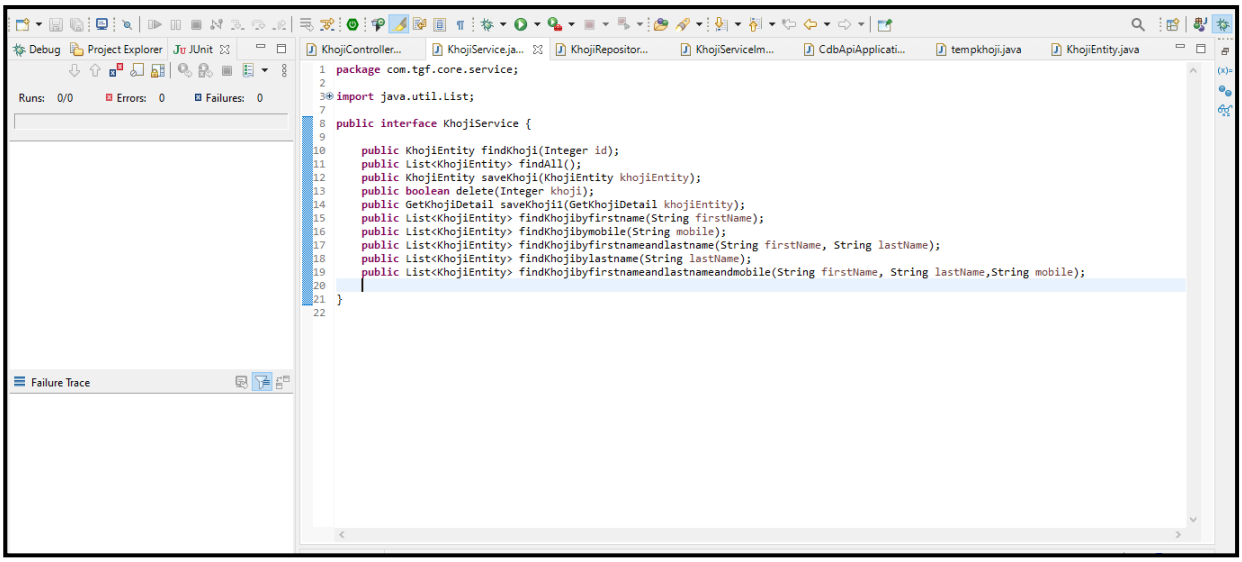


Fig 6.3.2 API's Screenshot –Khoji service

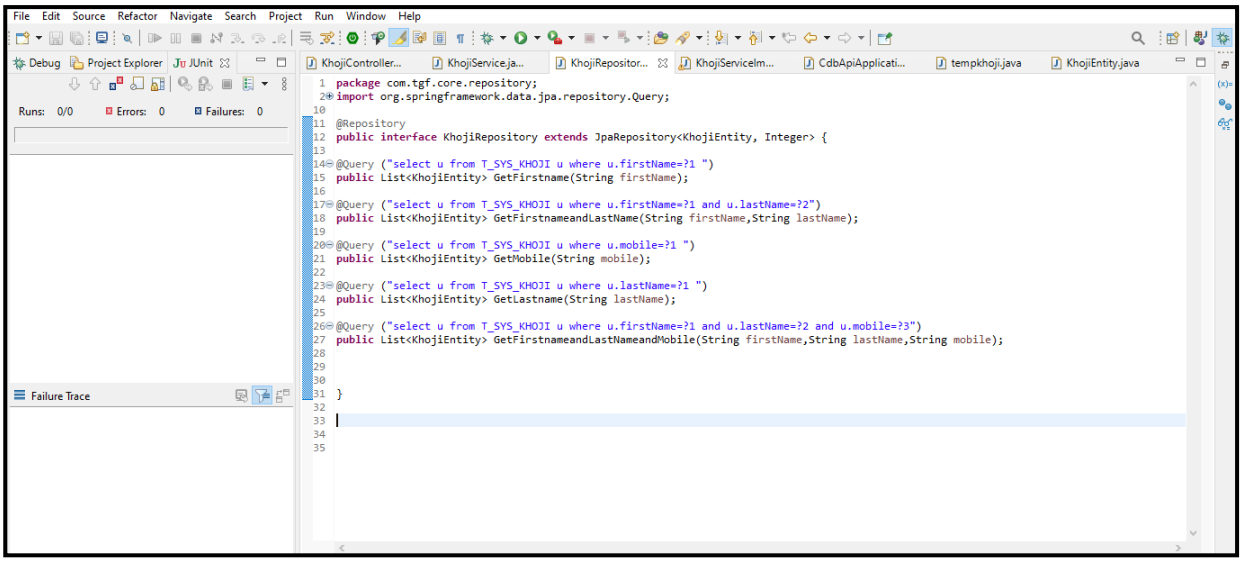


Fig 6.3.3 API's Screenshot – Search Khoji on the basis of first name ,last name

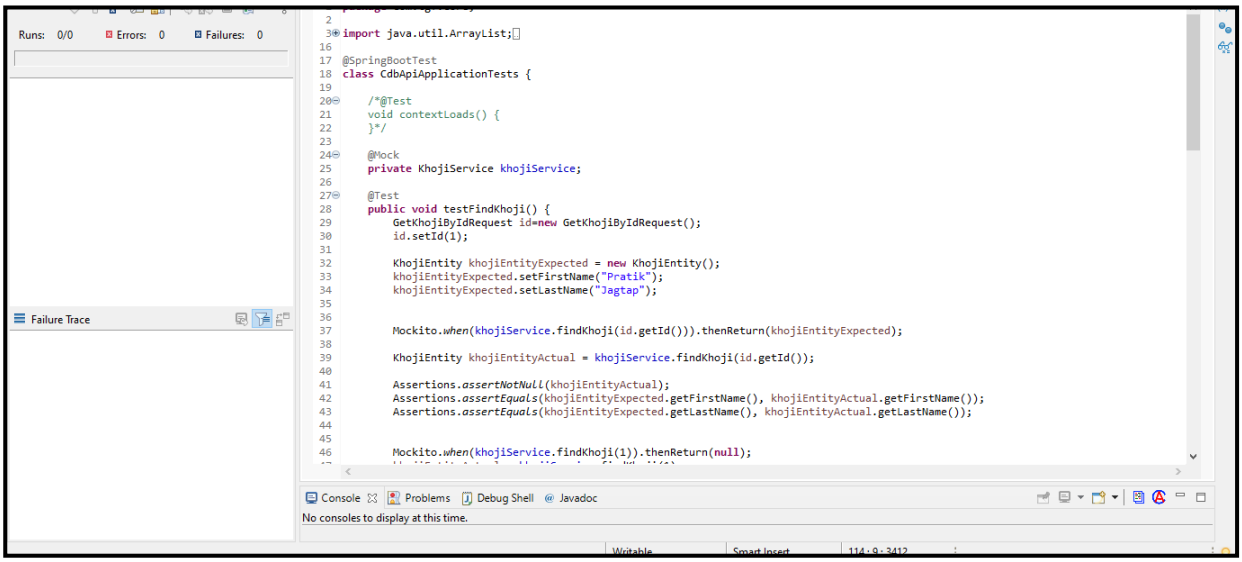


Fig 6.3.4 JUnit testing

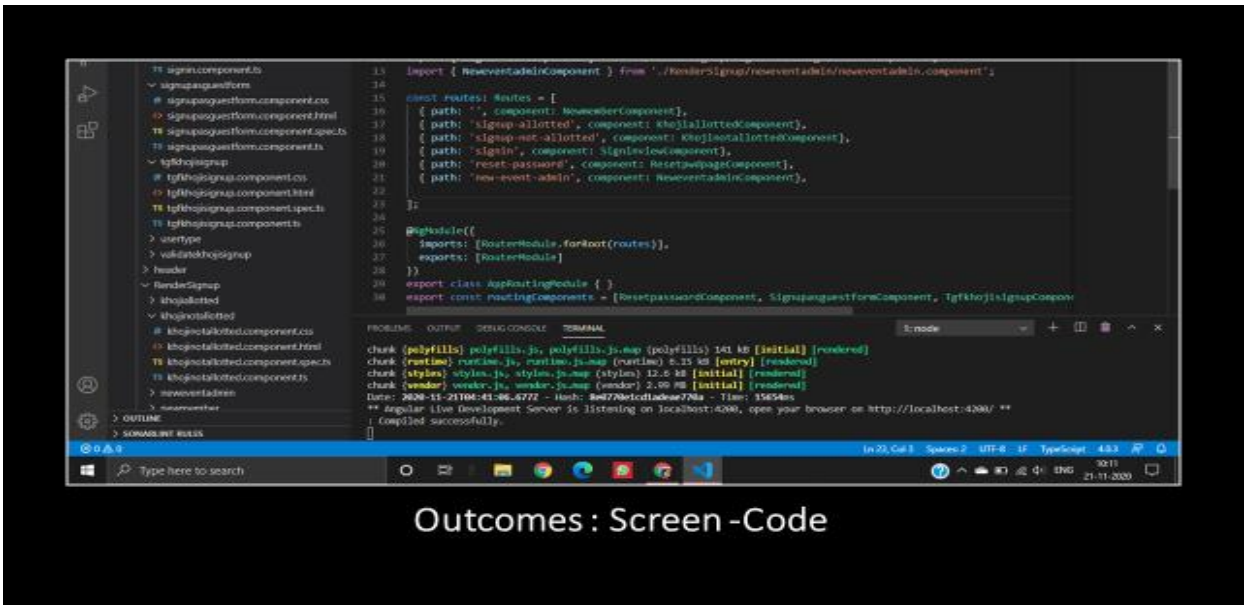


Fig 6.3.5 Angular

6.4 Integration of backend and frontend

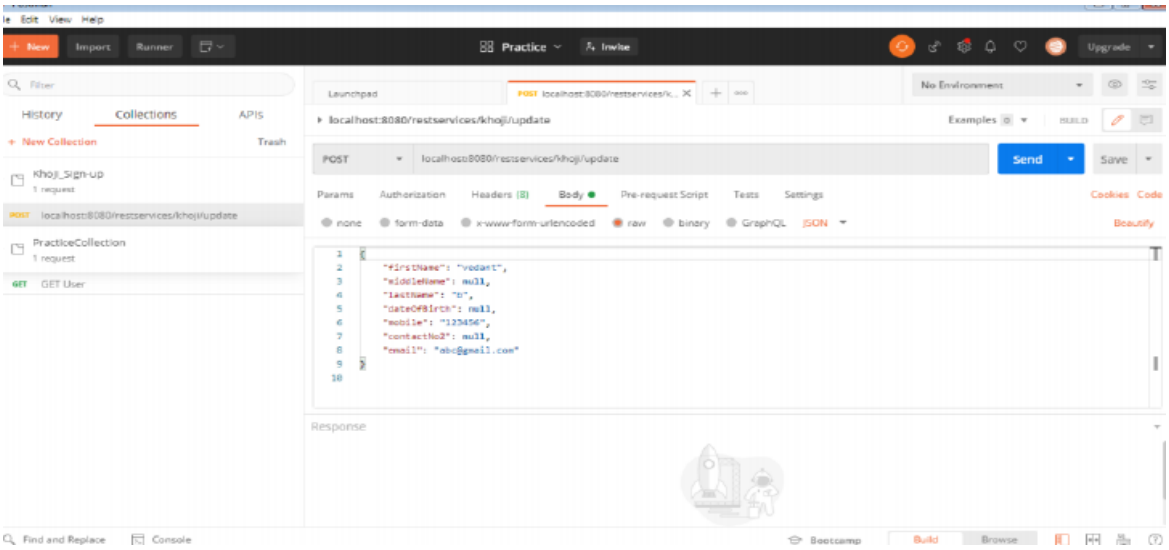


Fig 6.4.1 Integration Backend-Postman

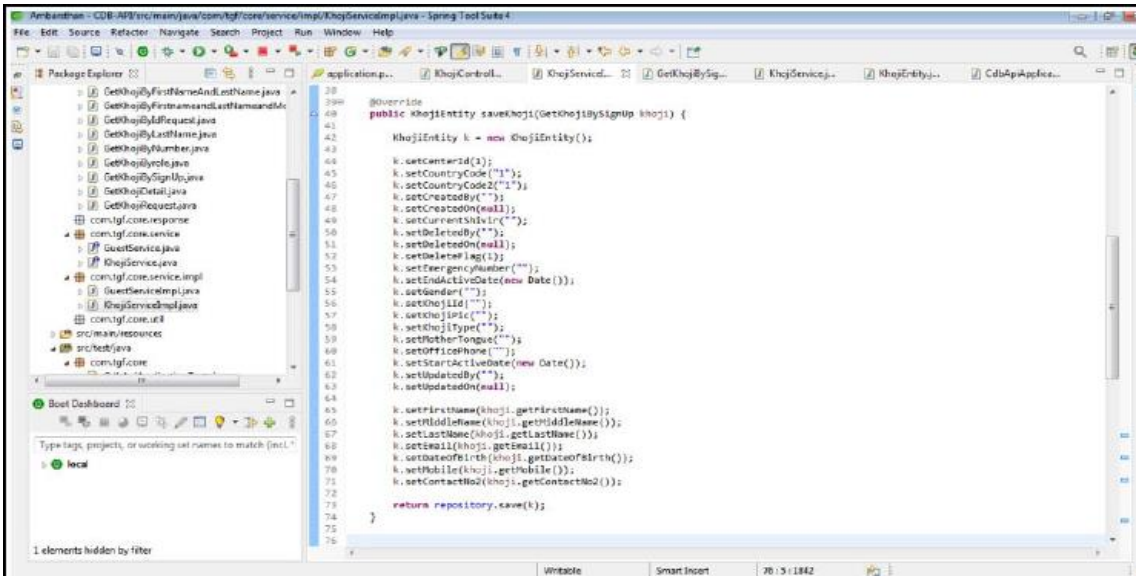


Fig 6.4.2 Integration Backend-Spring Suite

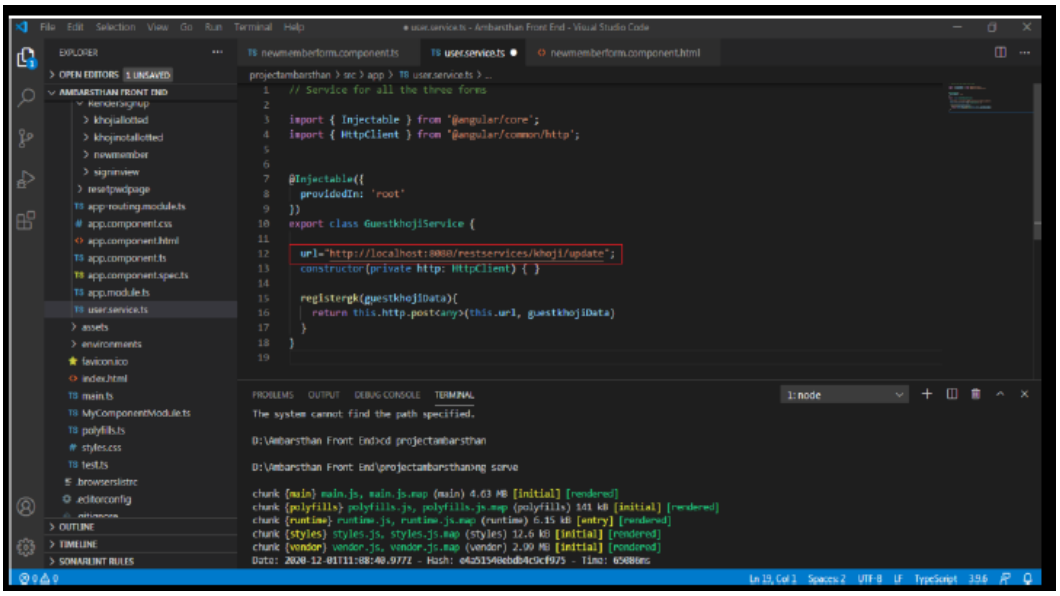


Fig 6.4.3 Integration Frontend-Visual Studio

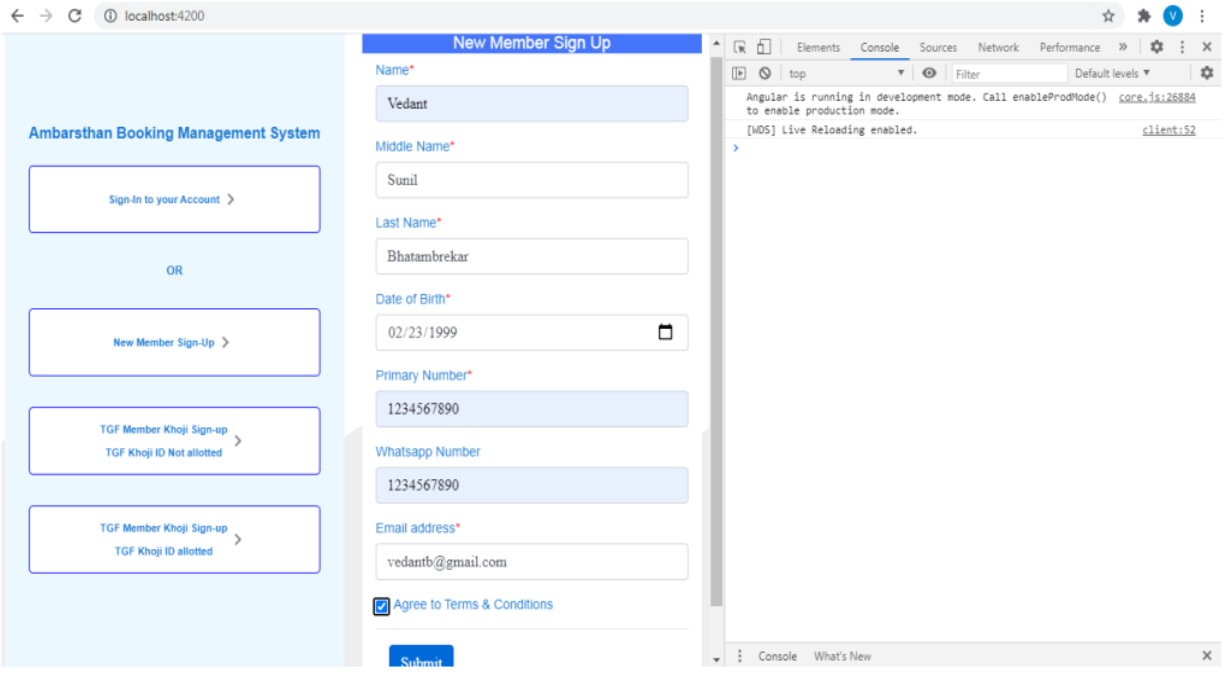


Fig 6.4.4 Integration Frontend-UI

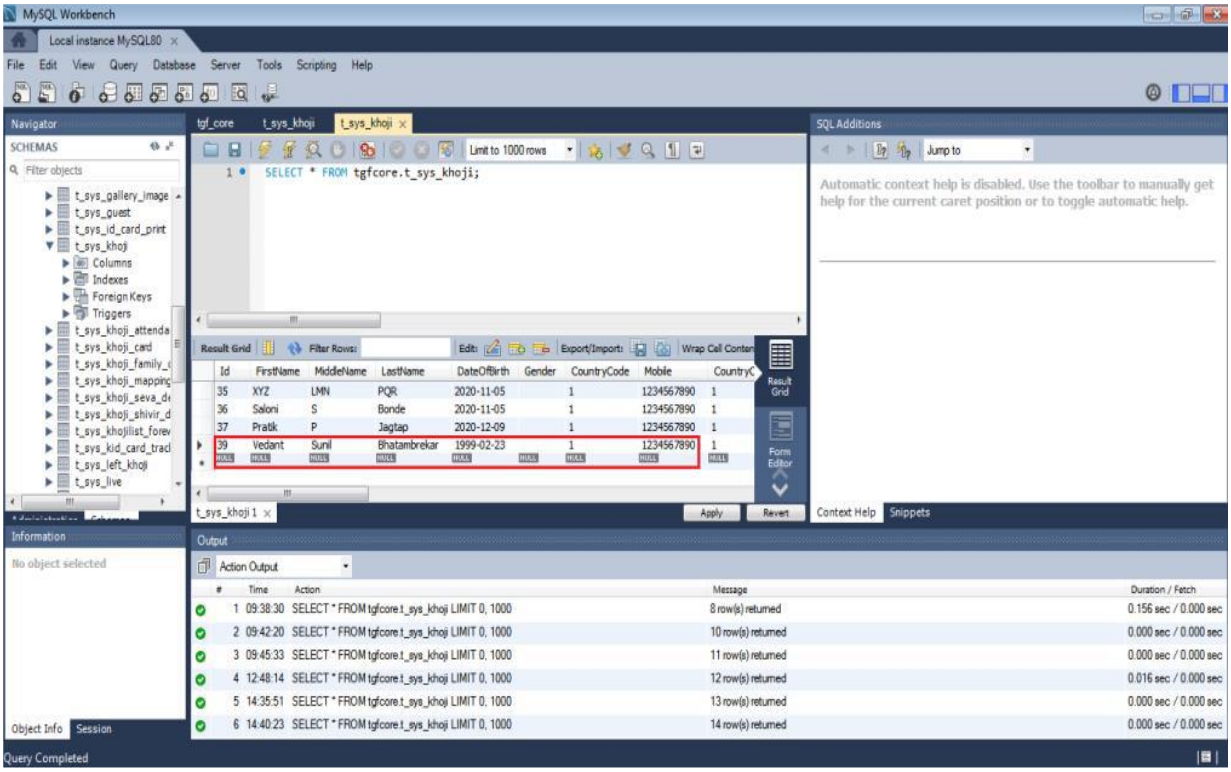


Fig 6.4.5 Integration –Tables Updated

6.5 Introduction

This document is a high-level overview defining our testing strategy for the website application. Its objective is to book events and accommodation for the respective event. It portrays a snapshot of the project as of the end of the planning phase. This document will address the different standards that will apply to the unit, integration and system testing of the specified application.

6.6 Purpose

The project aims at providing a user-friendly website for the booking of Ambarsthan complex of Tejgyan Foundation.In the Ambarsthan Accommodation Management System, we render booking services to the members of Tejgyan Foundation and give access/services to the admin to handle the database related to the complex.

6.7 Test Objective

The objective of our test plan is to find and report as many bugs as possible to improve

the integrity of our program. Although exhaustive testing is not possible, we will exercise a broad range of tests to achieve our goal. There will be five key functions used to manage our application: load, store, search, insert, delete. Our user interface to utilize these functions is designed to be user-friendly and provide easy manipulation of the data. The application will only be used as a demonstration tool, but we would like to ensure that it could be run from a variety of platforms with little impact on performance or usability.

6.8 Process Overview

The following represents the overall flow of the testing process:

1. Identify the requirements to be tested. All test cases shall be derived using the current Program Specification.
2. Identify which particular test will be used to test each module.
3. Review the test data and test cases to ensure that the unit has been thoroughly verified and that the test data and test cases are adequate to verify proper operation of the unit.
4. Identify the expected results for each test.
5. Document the test case configuration, test data, and expected results.
6. Perform the test.
7. Document the test data, test cases, and test configuration used during the testing process.
8. Successful JUnit testing is required before the unit is eligible for component integration/system testing.
9. Unsuccessful testing requires a Bug Report Form to be generated. This document shall describe the test case, the problem encountered, its possible cause, and the sequence of events that led to the problem. It shall be used as a basis for later technical analysis.
10. Test documents and reports shall be submitted. Any specifications to be reviewed, revised, or updated shall be handled immediately.

6.9 Test Cases And Results

Table No 6.1.4 Test Cases

Test Case No.	Test Case Name	Prerequisites	Action	Excepted Result	Actual Result	Status
1	Login	Username and password should be entered.	Click on login button	Successful login for correct username and password	Successful login	Pass
2	Login	Username and password should be entered.	Click on login button	Unsuccessful login for wrong username and password	Unsuccessful login	Pass
3	Signup	All details must be entered	Click on signup button	Successful sign-up for correct details	Successful Sign-up	Pass
4	Signup	All details must be entered	Click on signup button	Successful sign-up for correct details	Unsuccessful Sign-up for wrong details	Pass
5	Reset Password	Enter any value in input field	Click the button	Pop up of alert message	Entered value is wrong	Pass
6	Reset Password	Enter any value in input field	Click the button	Pop up of alert message	Entered value is correct	Pass

Chapter 7

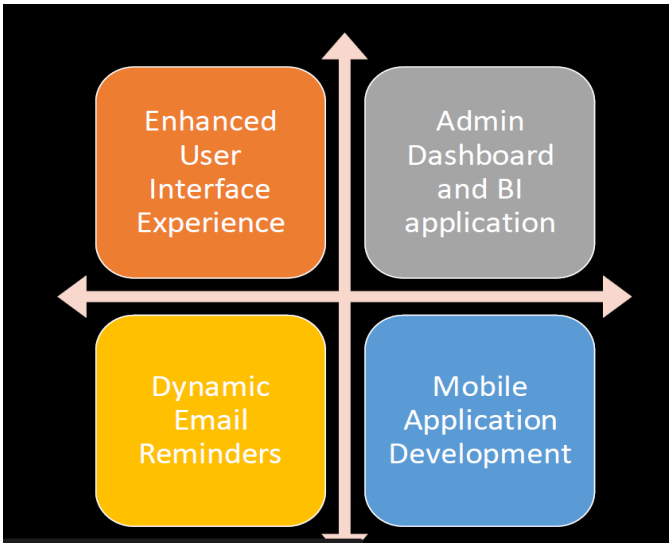
Conclusion and Future work

7.1 Conclusion

In this project we successfully implemented the tasks given to us. For the frontend, the team built Wireframes for the complete application using Framer Prototyping tool. This helped at finalizing the scope, developing business logic and gave an idea to the Product Owners. We were able to develop Screens for the wireframes provided to us by the TCS UI/UX team. At the backend, the team was able to build nearly 20+ APIs to implement the development logic and Capacity management logic as per the requirements. • We were also able to show PoC of Integration of Frontend and Backend.

7.2 Future Scope

In Ambarsthan Accommodation Management System, further modules can be designed as per the updates in scope, integration of the other modules in scope. Not only bookings but we can have reminders for upcoming events and get reminders for vacancies and manage multiple events with similar accommodation type and same unique credentials.



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